

Introduction to Bayesian Statistics

Part 4 Linear Models

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In this lecture

- What is a linear model?
- Continuous predictors (Regression)
- Categorical predictors (ANOVA)
- Categorical & continuous predictors (ANCOVA)

In-between:

- Model selection
- Post-hoc analysis

What is a linear model?

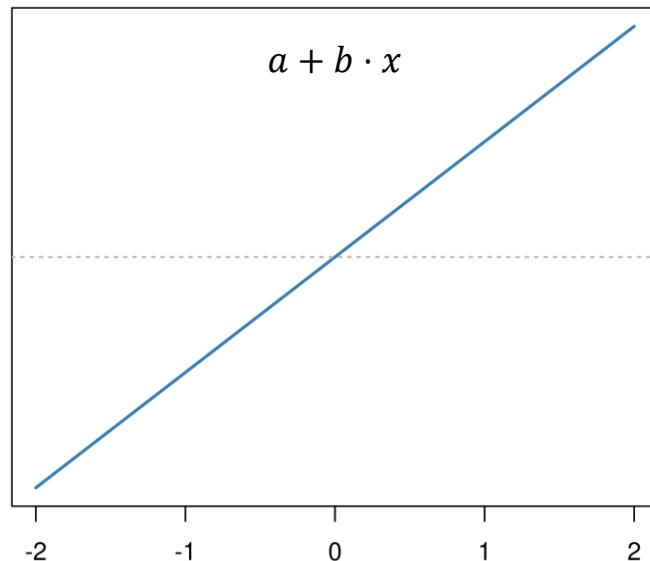
Linear functions

Linear in x (predictor)

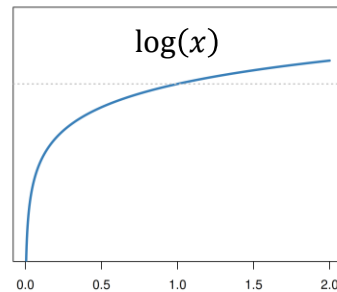
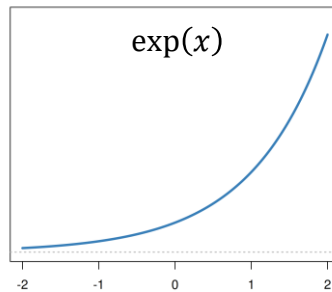
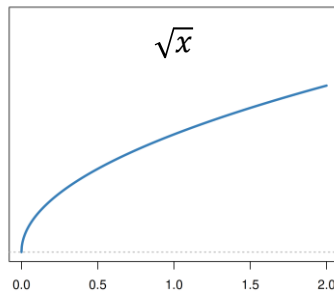
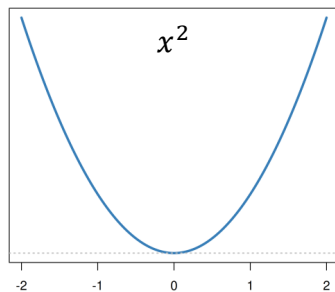
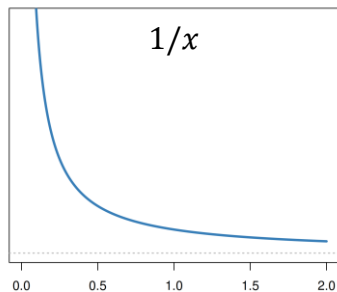
$$f(x) = a + b \cdot x$$

Additive with constant a (intercept)

Multiplication only with constant b (slope)



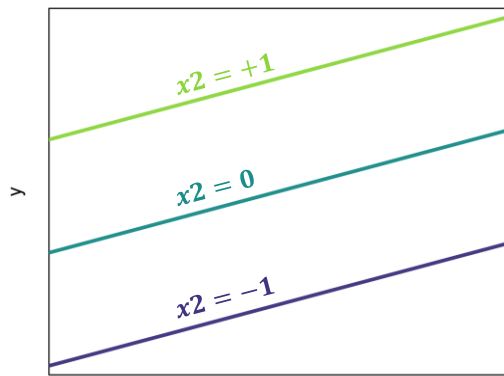
Some **nonlinear** functions:



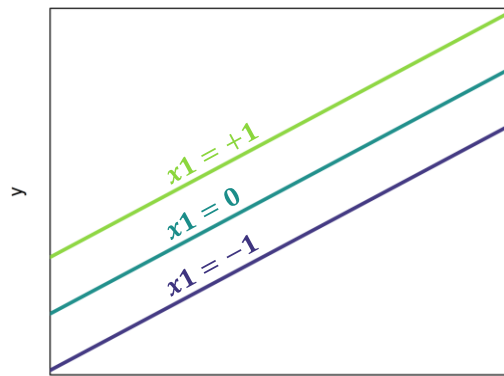
Linear functions

Extend to **multiple predictors** $x_1, x_2, \dots$

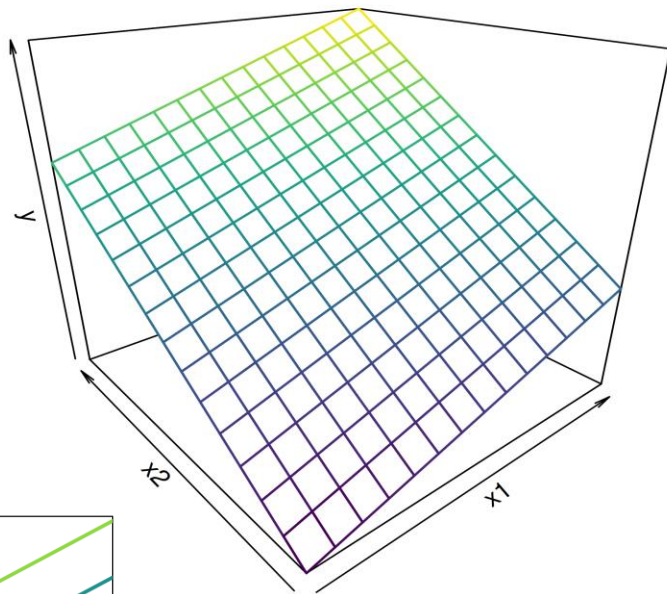
$$f(x) = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2$$



x_1



x_2



Linear statistical models

Linear in b (parameters) and
Gaussian random errors ε (normally distributed)

$$y(x) = b_0 + b_1 \cdot x + \varepsilon$$

$$y(x) = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + \varepsilon$$

Nonlinear in b , for example:

$$y(x) = b_0 + x^{b_1} + \varepsilon$$

$$y(x) = b_0 + \exp(b_1 \cdot x) + \varepsilon$$

Linear statistical models in the
frequentist world:

Analytical solution (formula) for
parameter estimates

Easy computation with `lm()`

Nonlinear models in the
frequentist world:

Maximum likelihood estimation
(iterative algorithm)

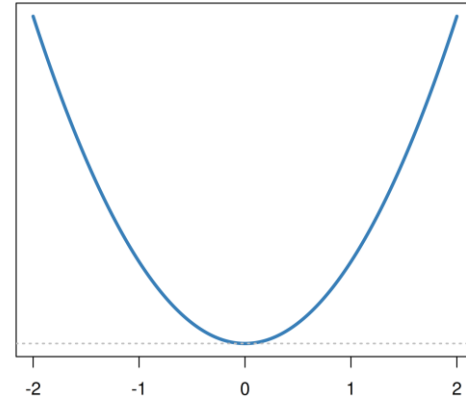
Linear statistical models

Quadratic (polynomial) relationships

$$y = b_0 + b_1 \cdot x + b_2 \cdot x^2 + \varepsilon$$

$$= b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + \varepsilon$$

define
 $x_1 = x$
 $x_2 = x^2$

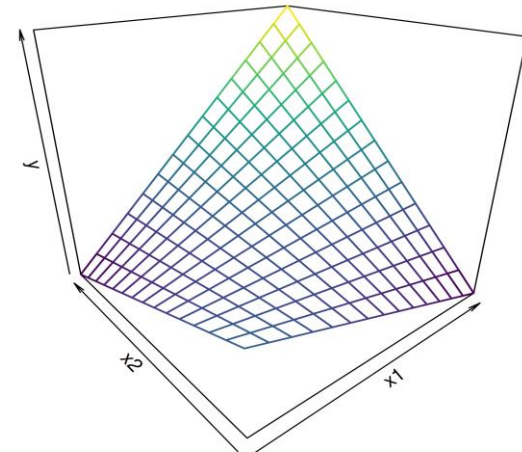


Interaction effects

$$y = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + b_3 \cdot x_1 \cdot x_2 + \varepsilon$$

$$= b_0 + b_1 \cdot x_1 + b_2 \cdot x_2 + b_3 \cdot x_3 + \varepsilon$$

define $x_3 = x_1 x_2$



→ Some nonlinear relationships can be described with linear statistical models (**linear in the parameters**)

Linear statistical models

Transformation of response variable

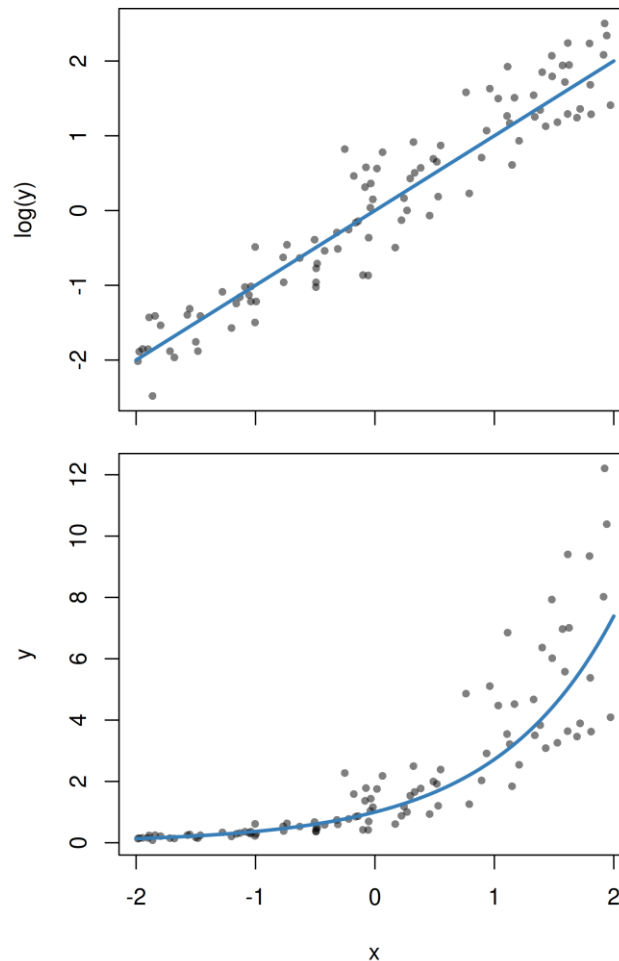
$$\log(y) = \mathbf{b}_0 + \mathbf{b}_1 \cdot x + \varepsilon$$

Attention: model becomes multiplicative

When back transforming to y -scale

$$\begin{aligned} y &= \exp(b_0 + b_1 \cdot x + \varepsilon) \\ &= \exp(b_0) \cdot \exp(b_1 x) \cdot \exp(\varepsilon) \\ &= \tilde{b}_0 \cdot \exp(b_1 x) \cdot \tilde{\varepsilon} \end{aligned}$$

→ Sometimes statistical models can be “linearized” by transformation



Common statistical tests are linear models

Last updated: 28 June, 2019. Also check out the [Python version!](#)

See worked examples and more details at the accompanying notebook: <https://lindeloef.github.io/tests-as-linear>

	Common name	Built-in function in R	Equivalent linear model in R	Exact?	The linear model in words	Icon
Simple regression: $\text{lm}(y \sim 1 + x)$	y is independent of x P: One-sample t-test N: Wilcoxon signed-rank	t.test(y) wilcox.test(y)	$\text{lm}(y \sim 1)$ $\text{lm}(\text{signed_rank}(y) \sim 1)$	✓ for N > 14	One number (intercept, i.e., the mean) predicts y. - (Same, but it predicts the <i>signed rank</i> of y.)	
	P: Paired-sample t-test N: Wilcoxon matched pairs	t.test(y1, y2, paired=TRUE) wilcox.test(y1, y2, paired=TRUE)	$\text{lm}(y_2 - y_1 \sim 1)$ $\text{lm}(\text{signed_rank}(y_2 - y_1) \sim 1)$	✓ for N > 14	One intercept predicts the pairwise $y_2 - y_1$ differences. - (Same, but it predicts the <i>signed rank</i> of $y_2 - y_1$.)	
	y ~ continuous x P: Pearson correlation N: Spearman correlation	cor.test(x, y, method='Pearson') cor.test(x, y, method='Spearman')	$\text{lm}(y \sim 1 + x)$ $\text{lm}(\text{rank}(y) \sim 1 + \text{rank}(x))$	✓ for N > 10	One intercept plus x multiplied by a number (slope) predicts y. - (Same, but with <i>ranked x</i> and y)	
	y ~ discrete x P: Two-sample t-test P: Welch's t-test N: Mann-Whitney U	t.test(y1, y2, var.equal=TRUE) t.test(y1, y2, var.equal=FALSE) wilcox.test(y1, y2)	$\text{lm}(y \sim 1 + G_2)^A$ $\text{gls}(y \sim 1 + G_2, \text{weights}=\dots^B)^A$ $\text{lm}(\text{signed_rank}(y) \sim 1 + G_2)^A$	✓ ✓ for N > 11	An intercept for group 1 (plus a difference if group 2) predicts y. - (Same, but with one variance <i>per group</i> instead of one common.) - (Same, but it predicts the <i>signed rank</i> of y.)	
Multiple regression: $\text{lm}(y \sim 1 + x_1 + x_2 + \dots)$	P: One-way ANOVA N: Kruskal-Wallis	aov(y ~ group) kruskal.test(y ~ group)	$\text{lm}(y \sim 1 + G_2 + G_3 + \dots + G_N)^A$ $\text{lm}(\text{rank}(y) \sim 1 + G_2 + G_3 + \dots + G_N)^A$	✓ for N > 11	An intercept for group 1 (plus a difference if group ≠ 1) predicts y. - (Same, but it predicts the <i>rank</i> of y.)	
	P: One-way ANCOVA	aov(y ~ group + x)	$\text{lm}(y \sim 1 + G_2 + G_3 + \dots + G_N + x)^A$	✓	- (Same, but plus a slope on x.) <i>Note: this is discrete AND continuous. ANCOVAs are ANOVAs with a continuous x.</i>	
	P: Two-way ANOVA	aov(y ~ group * sex)	$\text{lm}(y \sim 1 + G_2 + G_3 + \dots + G_N + S_2 + S_3 + \dots + S_K + G_2 * S_2 + G_3 * S_3 + \dots + G_N * S_K)^A$	✓	Interaction term: changing sex changes the y ~ group parameters. <i>Note: $G_{2:N,K}$ is an indicator (0 or 1) for each non-intercept levels of the group variable. Similarly for $S_{2:N,K}$ for sex. The first line (with G) is main effect of group, the second (with S) for sex and the third is the group * sex interaction. For two levels (e.g. male/female), line 2 would just be "S2" and line 3 would be S2 multiplied with each G.</i>	[Coming]
	Counts ~ discrete x N: Chi-square test	chisq.test(groupXsex_table)	Equivalent log-linear model $\text{glm}(y \sim 1 + G_2 + G_3 + \dots + G_N + S_2 + S_3 + \dots + S_K + G_2 * S_2 + G_3 * S_3 + \dots + G_N * S_K, \text{family}=\dots)^A$	✓	Interaction term: (Same as Two-way ANOVA.) <i>Note: Run glm using the following arguments: <code>glm(model, family=poisson())</code> As linear-model, the Chi-square test is $\log(y) = \log(N) + \log(a) + \log(\beta) + \log(a\beta)$ where a and β are proportions. See more info in the accompanying notebook.</i>	Same as Two-way ANOVA
	N: Goodness of fit	chisq.test(y)	$\text{glm}(y \sim 1 + G_2 + G_3 + \dots + G_N, \text{family}=\dots)^A$	✓	(Same as One-way ANOVA and see Chi-Square note.)	1W-ANOVA

<https://lindeloef.github.io/tests-as-linear/>

Bayesian stats & linearity

Linearity actually **not that important**

MCMC does not care if deterministic model part is

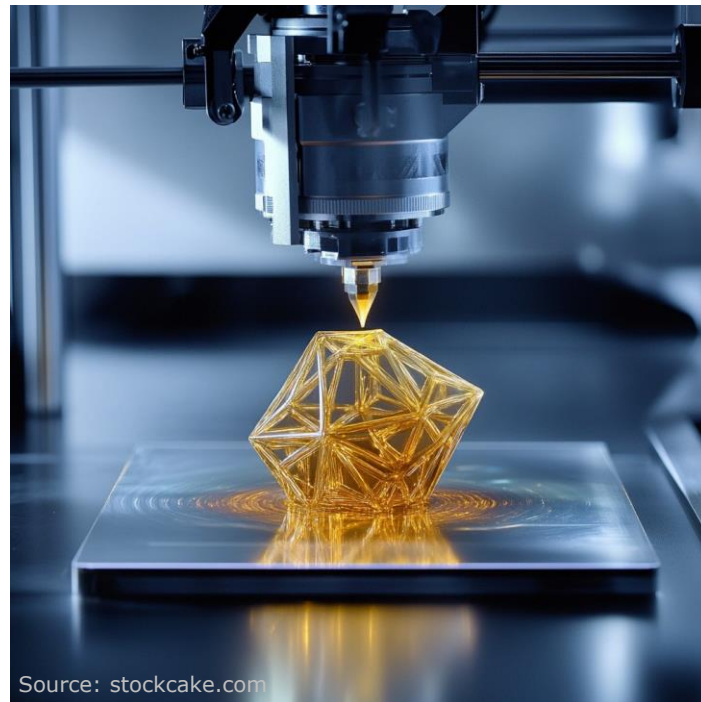
linear $\mu(x) = a + b \cdot x$

or nonlinear $\mu(x) = \frac{a \cdot x}{b + x}$

However, nonlinear (and also polynomial) models should only be considered when there is a good reason, not just because they would fit the data better.

Principle of parsimony, Occam's razor (14th century):
"Entities must not be multiplied beyond necessity"

The Bayesian 3D printer



Source: stockcake.com

*Continuous predictors
(Linear regression)*

Single predictor

Example: Latitudinal gradient of plant size

Global database with:

- $\log_{10}$ of plant height as response
- latitude as predictor

Later: include precipitation as environmental predictor



Single predictor

Example: Latitudinal gradient of plant size

Stochastic part: $\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = b_0 + b_1 \cdot \text{lat}$

```
> brm(log(height)~lat, data=globalPlants)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: log(height) ~ lat
Data: globalPlants (Number of observations: 131)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

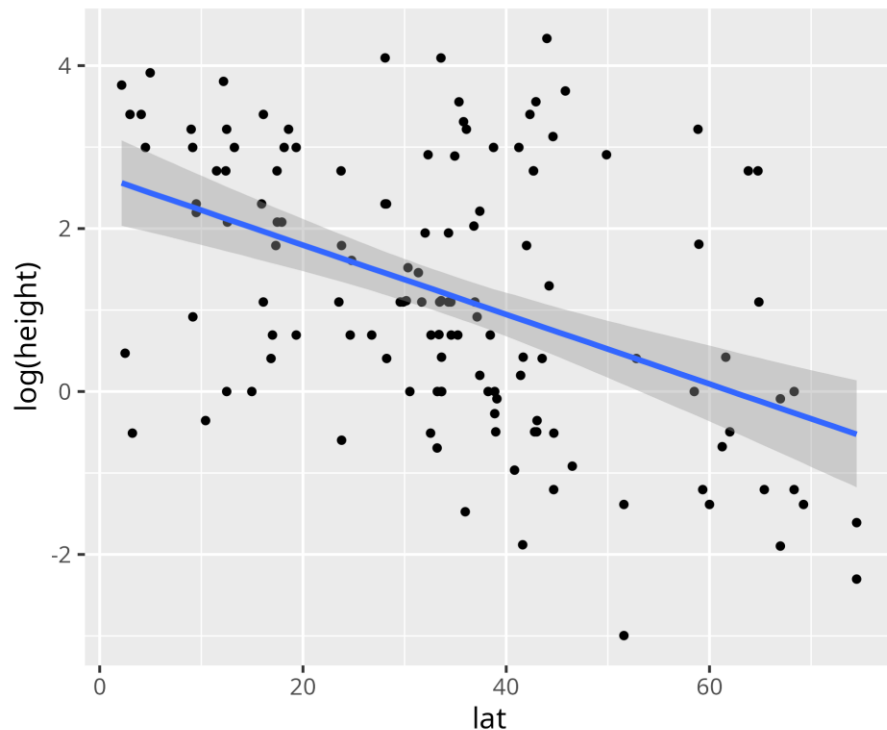
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	2.65	0.28	2.09	3.20	1.00	3634	2805
lat	-0.04	0.01	-0.06	-0.03	1.00	4126	2981

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	1.50	0.09	1.33	1.70	1.00	3419	2835

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).



Single predictor

Example: Latitudinal gradient of plant size

Stochastic part: $\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = b_0 + b_1 \cdot \text{lat}$

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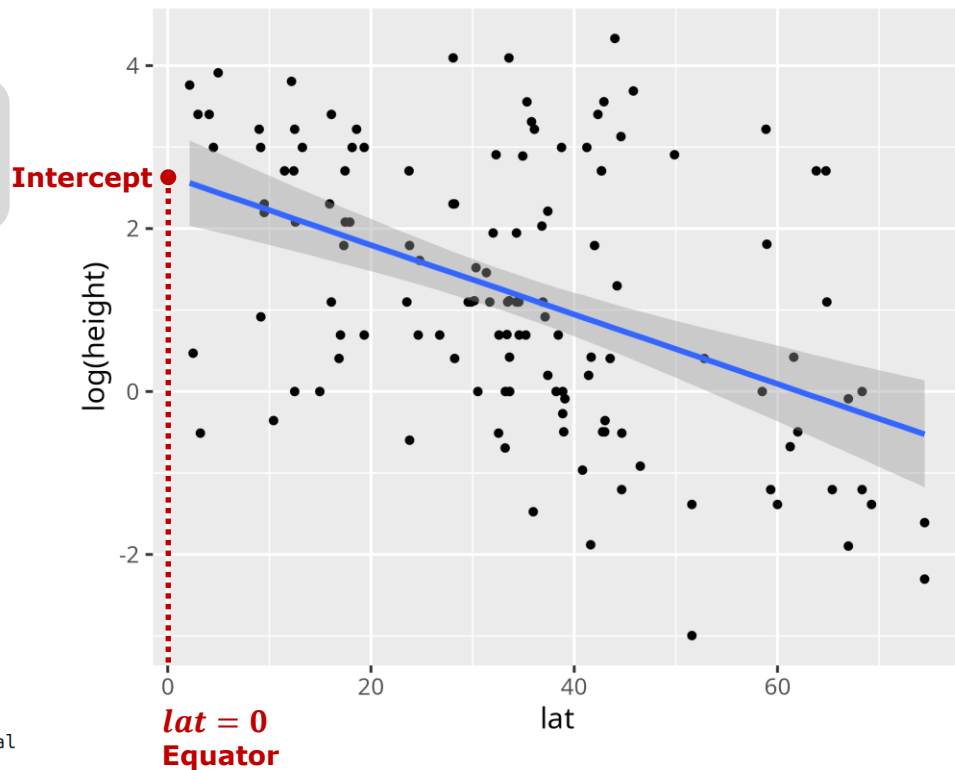
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Single predictor

Example: Latitudinal gradient of plant size

scale predictor ($\rightarrow$ mean = 0, sdev = 1)

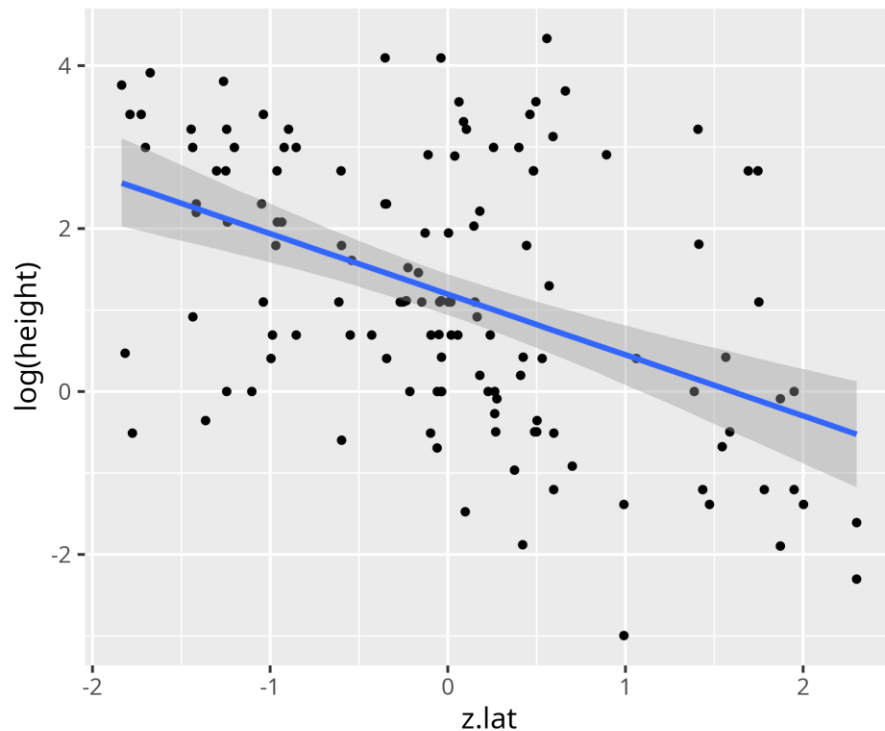
$$z.lat = \frac{lat - \text{mean}(lat)}{\text{sdev}(lat)} \text{ „Z-score“}$$

Deterministic part: $\mu = b_0 + b_1 \cdot z.lat$

```
> brm(log(height)~z.lat)
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.19	0.13	0.94	1.44	1.00	3799	2873
z.lat	-0.75	0.13	-1.00	-0.49	1.00	4196	2815



Single predictor

Example: Latitudinal gradient of plant size

scale predictor ($\rightarrow$ mean = 0, sdev = 1)

$$z.lat = \frac{lat - \text{mean}(lat)}{\text{sdev}(lat)} \text{ „Z-score“}$$

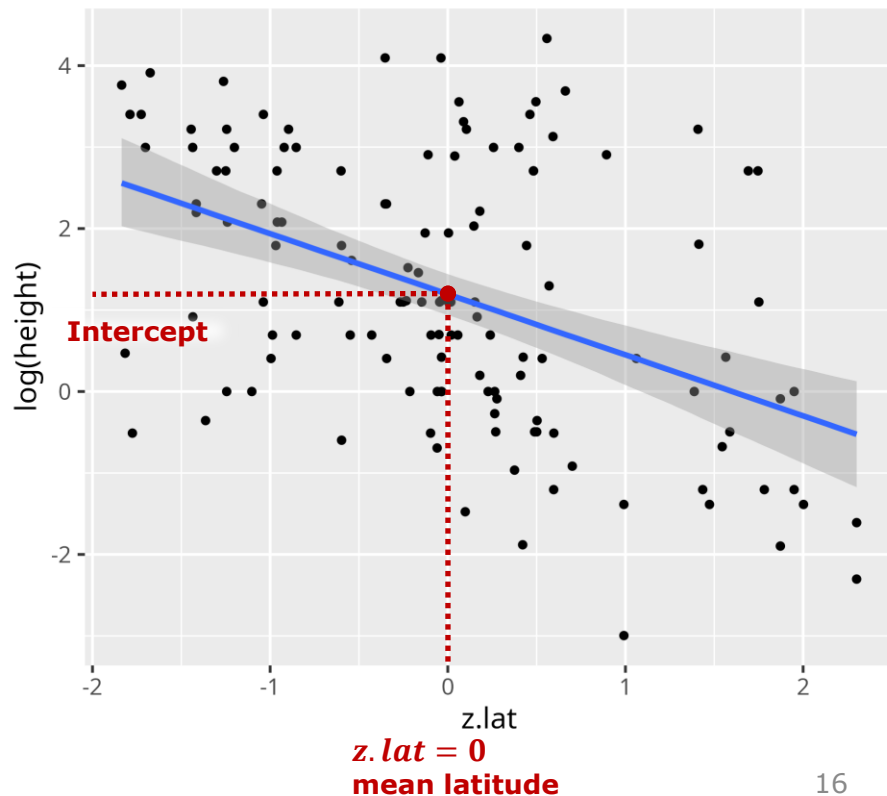
Deterministic part: $\mu = b_0 + b_1 \cdot z.lat$

```
> brm(log(height)~z.lat)
```

Regression Coefficients:

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Intercept	1.19	0.13	0.94	1.44	1.00	3799	2873
z.lat	-0.75	0.13	-1.00	-0.49	1.00	4196	2815

Now intercept is predicted $\log(\text{height})$
when predictor lat is at its average and
slope is effect for 1 sdev increment of lat



Multiple predictors

Example: Latitudinal gradient of plant size

Stochastic part: $\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = b_0 + b_1 \cdot \text{lat} + b_2 \cdot \text{rain}$

```
> brm(log(height)~z.lat+z.rain)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: log(height) ~ z.lat + z.rain
Data: globalPlants (Number of observations: 131)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
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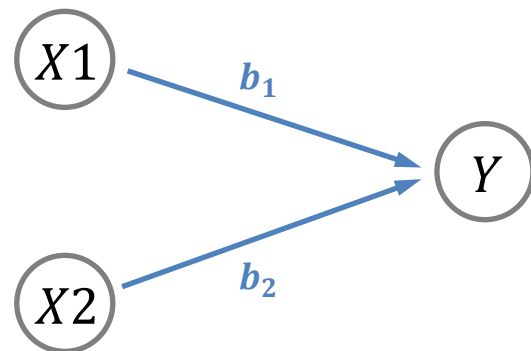
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.20	0.13	0.95	1.44	1.00	3186	2563
z.lat	-0.48	0.16	-0.78	-0.19	1.00	3490	3236
z.rain	0.46	0.15	0.16	0.76	1.00	3518	3088

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	1.45	0.09	1.29	1.65	1.00	3605	2835

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).



Multiple predictors

Example: Latitudinal gradient of plant size

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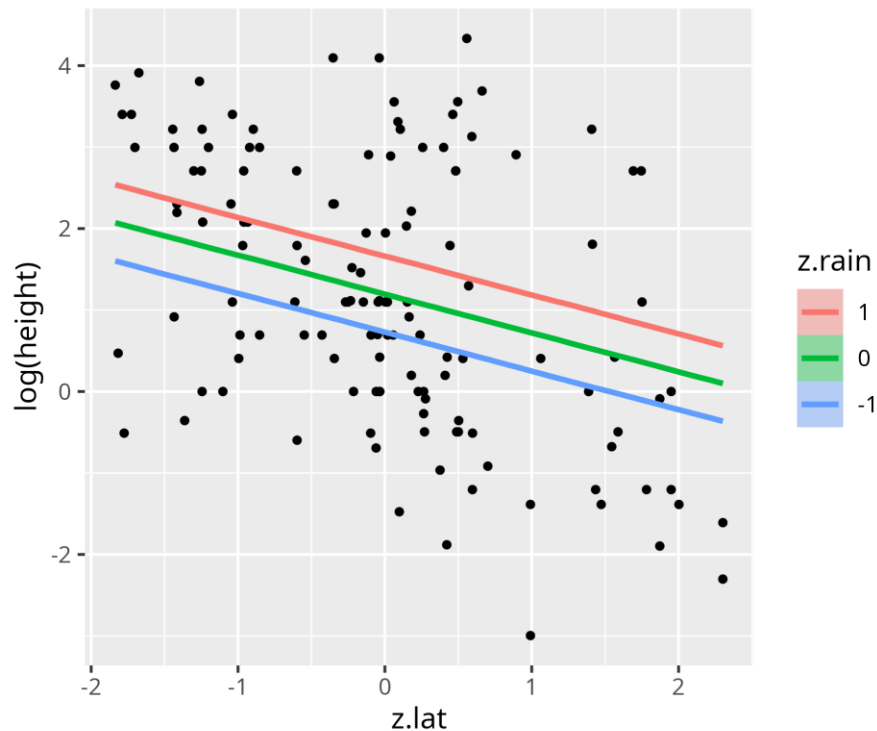
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Multiple predictors

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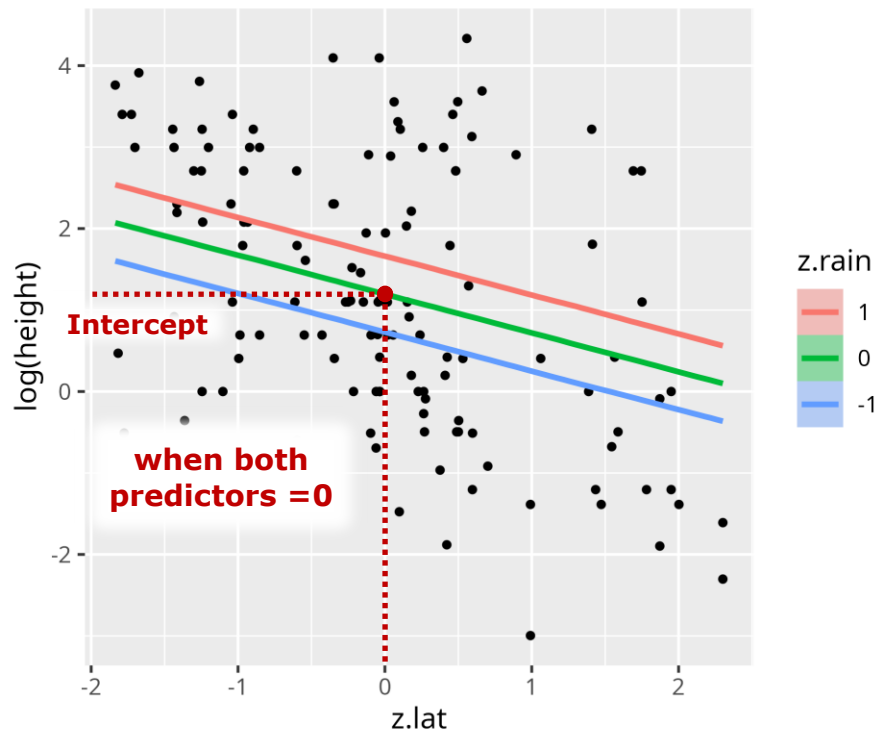
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Further Distributional Parameters:

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Multiple predictors

Example: Latitudinal gradient of plant size

Stochastic part: $\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = b_0 + b_1 \cdot \text{lat} + b_2 \cdot \text{rain}$

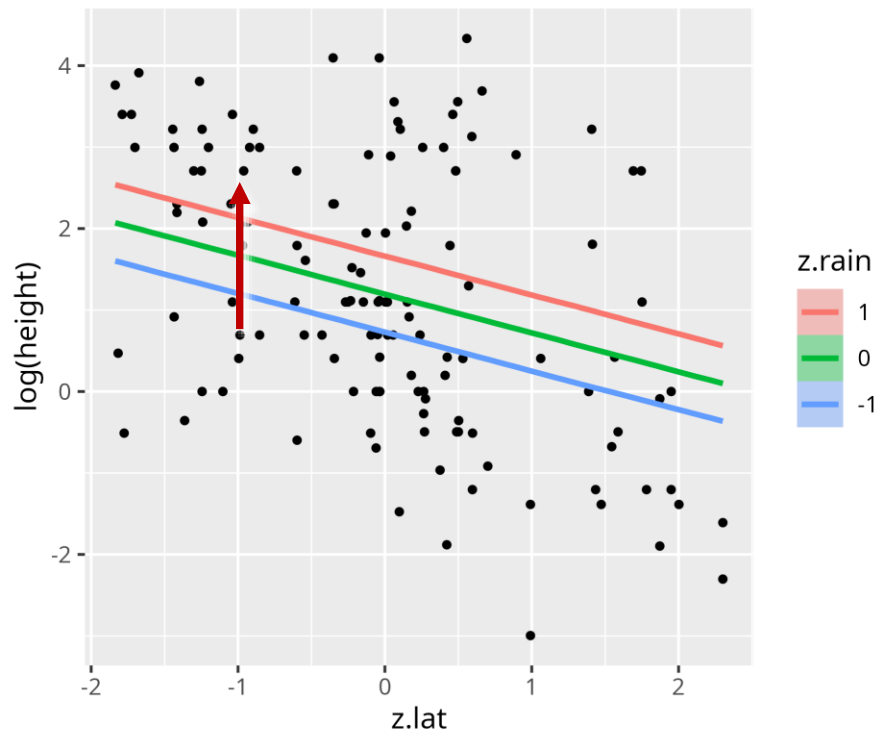
$$\mu = (b_0 + \mathbf{b_2} \cdot \text{rain}) + b_1 \cdot \text{lat}$$

Intercept depends
on **rain**

Slope
constant

2nd variable shifts intercept by b_2

Simpler interpretation when using
scaled variables $z.\text{lat}$ and $z.\text{rain}$



Multiple predictors

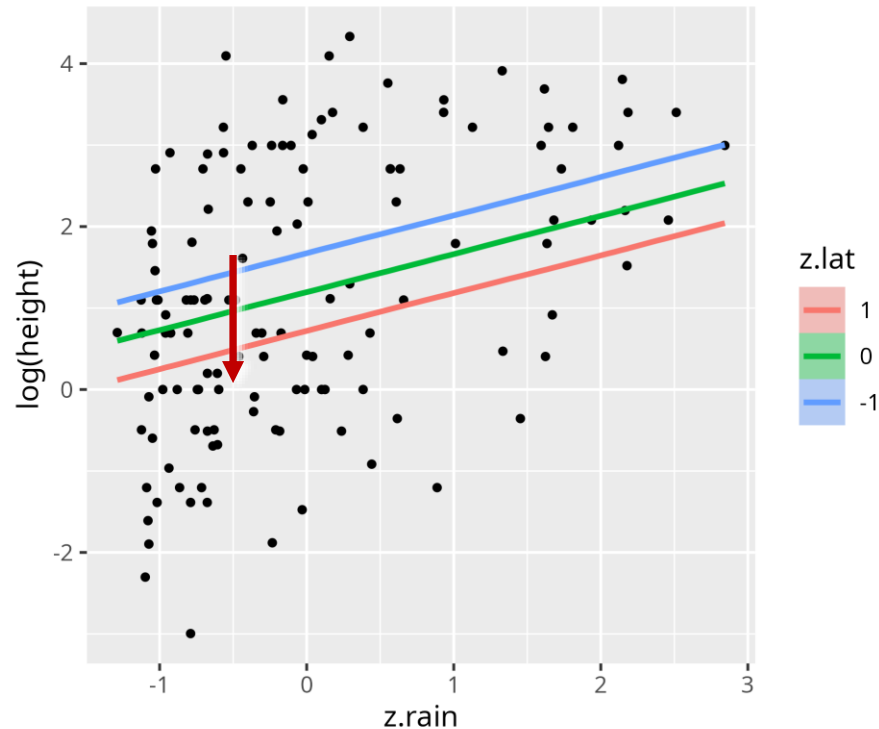
Now let's look at it from the perspective
the 2nd predictor *rain*

$$\mu = (b_0 + \mathbf{b_1} \cdot \mathbf{lat}) + b_2 \cdot \mathbf{rain}$$

Intercept depends
on *lat*

Slope
constant

1st variable shifts intercept by b_1



Multiple predictors: multicollinearity

Example: Latitudinal gradient of plant size

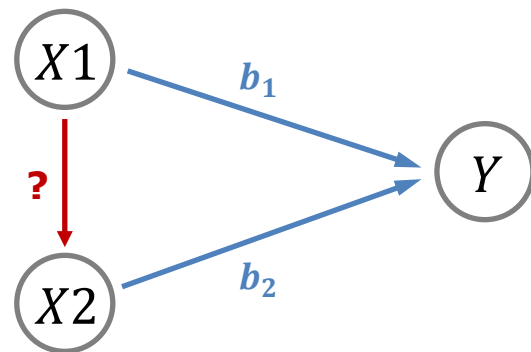
Stochastic part: $\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = b_0 + b_1 \cdot \text{lat} + b_2 \cdot \text{rain}$

What if predictor variables are correlated ?

Here *lat* influences *rain* !

- A bit of multicollinearity is OK.
- But be aware of interpretation of effects!
- b_1 is effect $x_1 \rightarrow y$, while x_2 held constant!
- Slopes describe direct (isolated) effects only, not total effect
- Often the problem when dealing with observational data instead controlled experiments.



→ Cinelli, Forney & Pearl (2024). A crash course in good and bad controls

Multiple predictors & interaction

Example: Latitudinal gradient of plant size

$\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$

$\mu = b_0 + b_1 \cdot \text{lat} + b_2 \cdot \text{rain} + \mathbf{b_3 \cdot \text{lat} \cdot \text{rain}}$

```
> brm(log(height)~z.lat*z.rain)
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Family: gaussian
Links: mu = identity; sigma = identity
Formula: log(height) ~ z.lat * z.rain
Data: globalPlants (Number of observations: 131)
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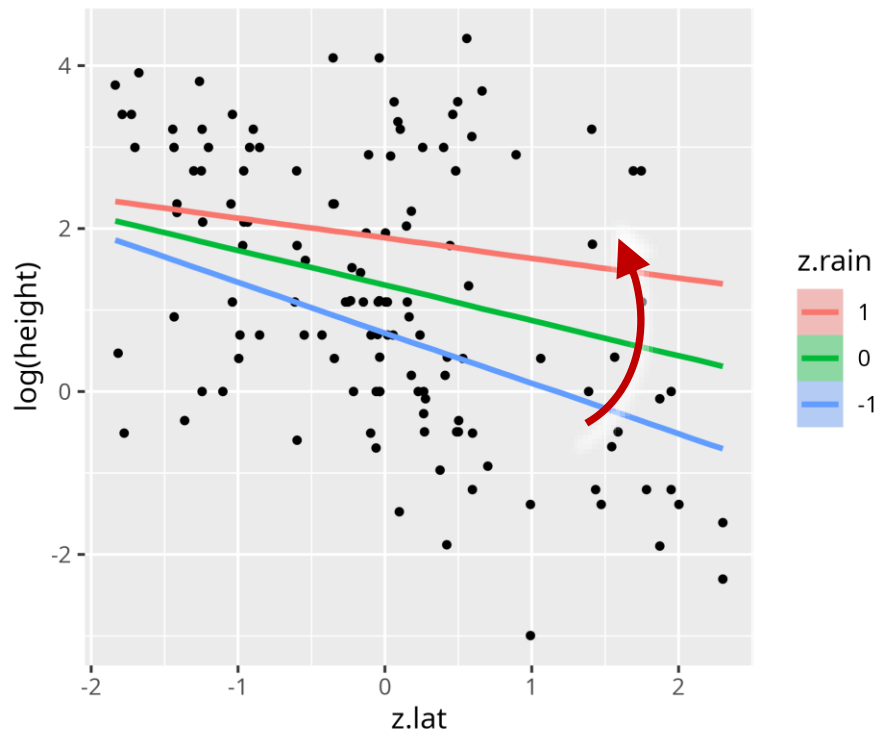
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.30	0.15	1.00	1.59	1.00	4804	2827
z.lat	-0.43	0.16	-0.74	-0.11	1.00	3606	3169
z.rain	0.58	0.18	0.23	0.95	1.00	3153	3169
z.lat:z.rain	0.19	0.14	-0.09	0.47	1.00	4010	2982

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	1.45	0.09	1.28	1.64	1.00	4778	2801

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).



Multiple predictors & interaction

Example: Latitudinal gradient of plant size

$$\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$$

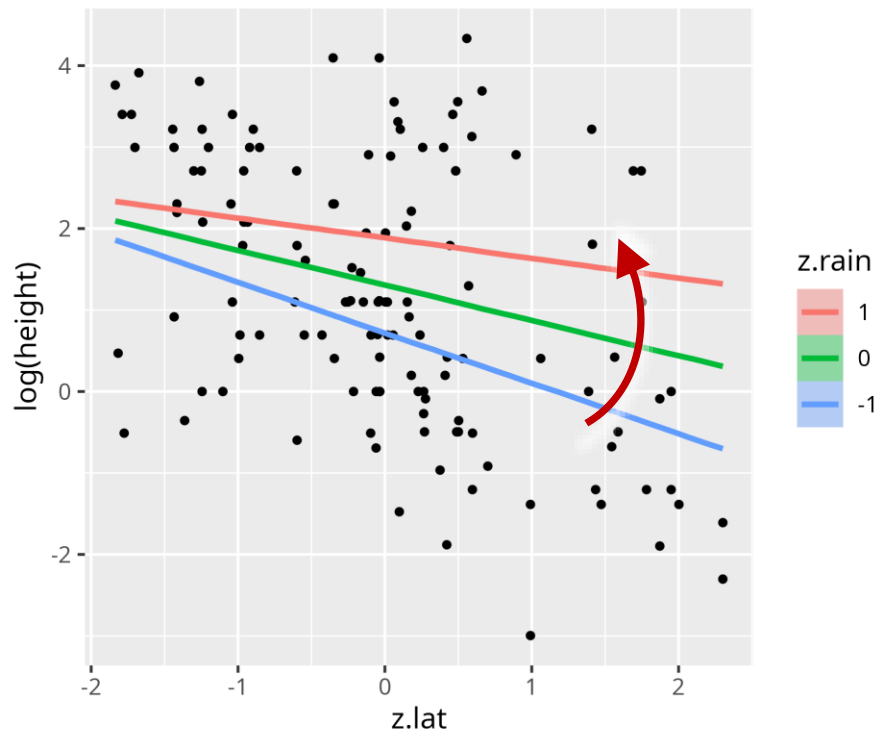
$$\mu = b_0 + b_1 \cdot \text{lat} + b_2 \cdot \text{rain} + \mathbf{b_3 \cdot lat \cdot rain}$$

$$\mu = (b_0 + b_2 \cdot \text{rain}) + (b_1 + \mathbf{b_3 \cdot rain}) \cdot \text{lat}$$

Intercept depends
on **rain**

Slope also depends
on **rain**

2nd variable shifts intercept by b_2
shifts slope by b_3



Multiple predictors & interaction

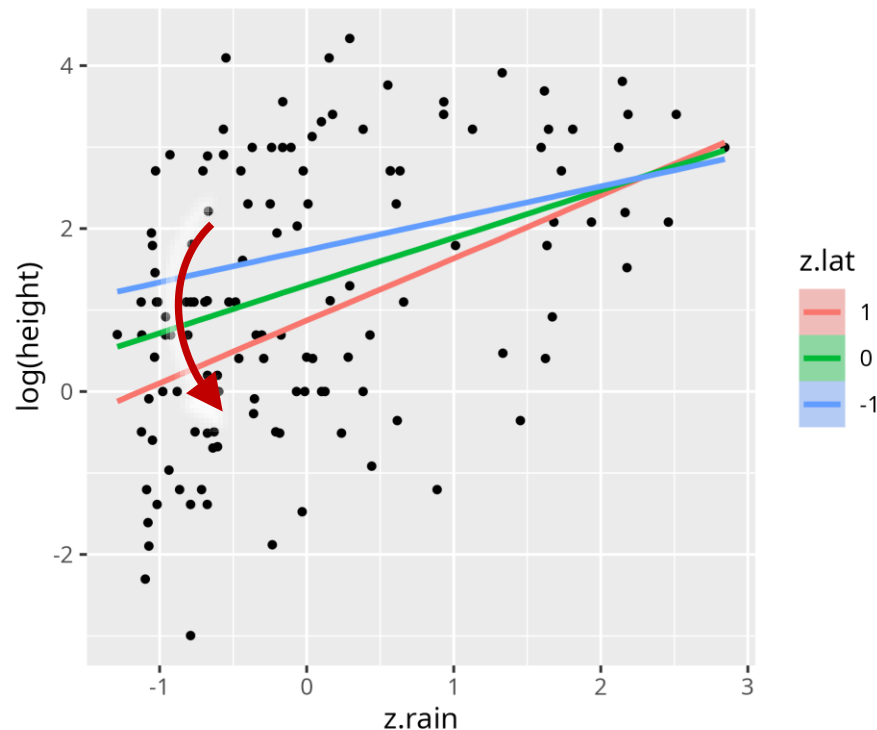
Now let's look at it from the perspective
the 2nd predictor *rain*

$$\mu = (b_0 + b_1 \cdot \text{lat}) + (b_2 + \mathbf{b_3} \cdot \text{lat}) \cdot \text{rain}$$

Intercept depends
on *lat*

Slope also depends
on *lat*

2nd variable shifts intercept by b_2
shifts slope by b_3



Multiple predictors & interaction

$$\log(\text{height}) \sim \text{Normal}(\mu, \sigma)$$

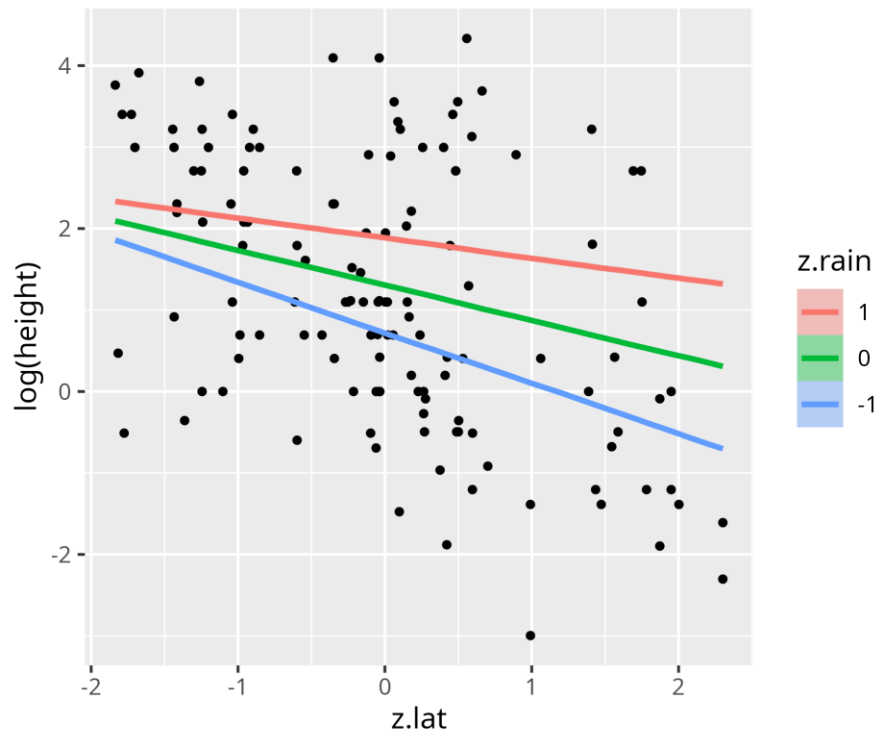
$$\mu = b_0 + b_1 \cdot \text{lat} + b_2 \cdot \text{rain} + b_3 \cdot \text{lat} \cdot \text{rain}$$

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	1.30	0.15	1.00	1.59	1.00	4804	2827
z.lat	-0.43	0.16	-0.74	-0.11	1.00	3606	3169
z.rain	0.58	0.18	0.23	0.95	1.00	3153	3169
z.lat:z.rain	0.19	0.14	-0.09	0.47	1.00	4010	2982

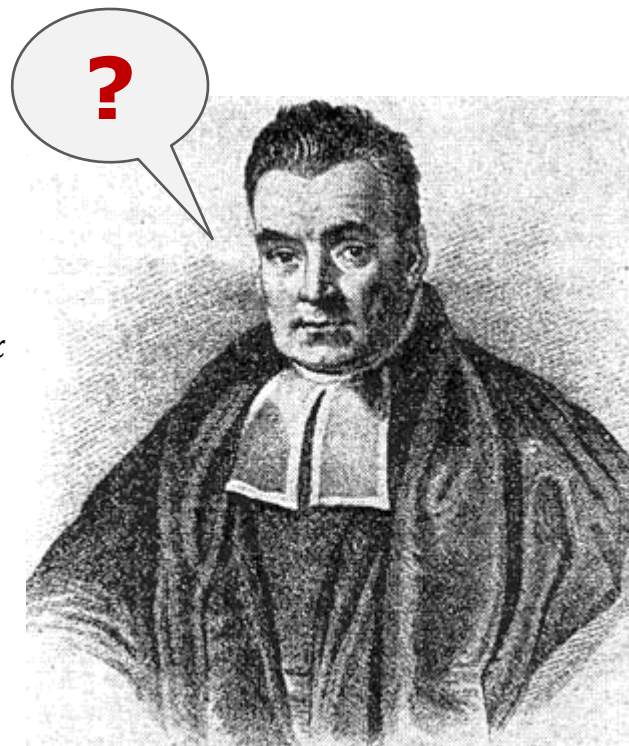
„Main effects“ describe slope of a predictor, when other predictors = 0!

Much simpler interpretation when using **scaled** variables *z.lat* and *z.rain*



Bayesian stats & linear regression

- Simple solutions for violation of model assumptions
 - Outliers → student-t distribution for residuals (heavier tails)
 - Non-constant residual sdev? → distributional models $\sigma(x) = \sigma_0 + \sigma_1 x$
 - Spatially / temporally autocorrelated residuals
- Simple comparison of intercepts & slopes („post-hoc analysis“)
- Regularization of effect sizes with priors
- Unbiased estimates even for small datasets
- Multivariate extensions (fit multiple responses at once)
- ...



Model selection

Frequentist F-tests

F-tests for (nested) linear models

Compare sums-of-squares of residuals

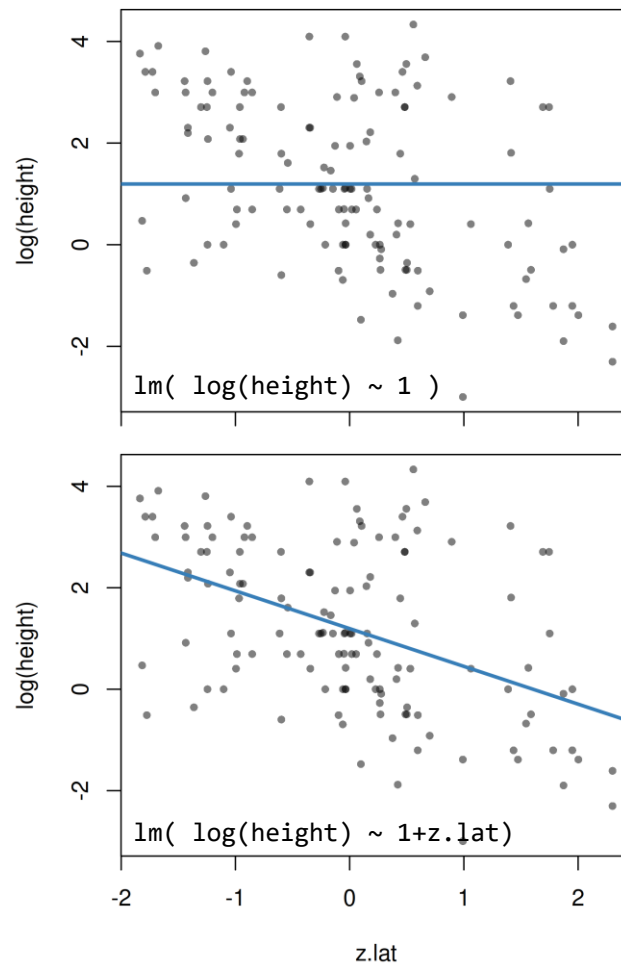
→ Connected to **R²** values (amount of explained variation)

R² always increases when adding predictors

H₀: Both models perform equally

F-test checks if increase is „significant“ or just random

$P < 0.05$ → reject H₀ and accept more complex model



Frequentist F-tests

```
lm( log(height) ~ z.lat*z.rain )
```

```
> summary( lm1 )
```

Call:

```
lm(formula = log(height) ~ z.lat * z.rain, data = globalPlants)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.2619	-0.9048	0.0017	1.0176	3.0977

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.3031	0.1490	8.745	1.14e-14 ***
z.lat	-0.4298	0.1581	-2.719	0.00746 **
z.rain	0.5855	0.1790	3.272	0.00138 **
z.lat:z.rain	0.1898	0.1411	1.345	0.18107

Don't use p-values of main effects
when there are higher-order effects
(here interaction) → **Use anova-table**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.435 on 127 degrees of freedom

Multiple R-squared: 0.2653, Adjusted R-squared: 0.2479

F-statistic: 15.29 on 3 and 127 DF, p-value: 1.5e-08

~ z.lat*z.rain versus ~ 1

→ Always tests full model against intercept-only

Frequentist F-tests

```
lm( log(height) ~ z.lat*z.rain )
```

```
> summary( lm1 )
```

Call:

```
lm(formula = log(height) ~ z.lat * z.rain, data = globalPlants)
```

Residuals:

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-3.2619	-0.9048	0.0017	1.0176	3.0977

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F-statistic: 15.29 on 3 and 127 DF, p-value: 1.5e-08

~ z.lat*z.rain **versus** ~ 1

→ Always tests full model against intercept-only

```
> anova( lm1 )
```

Analysis of Variance Table

Response: log(height)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
1: z.lat	1	72.083	72.083	35.0077	2.86e-08 ***
2: z.rain	1	18.612	18.612	9.0388	0.003186 **
3: z.lat:z.rain	1	3.724	3.724	1.8086	0.181073
Residuals	127	261.502	2.059		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

1: ~ z.lat **versus** ~ 1

2: ~ z.lat+z.rain **versus** ~ z.lat

3: ~ z.lat*z.rain **versus** ~ z.lat+z.rain

→ Incrementally tests more complex models

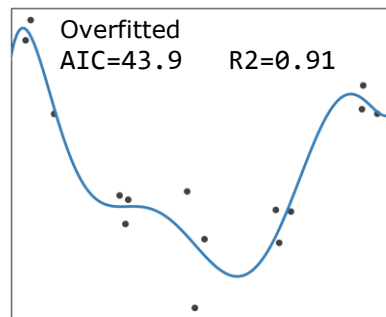
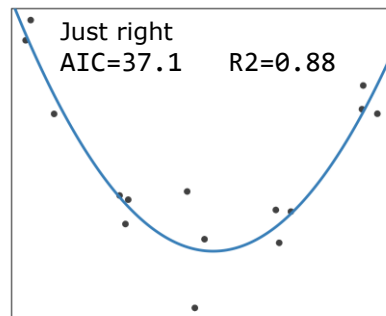
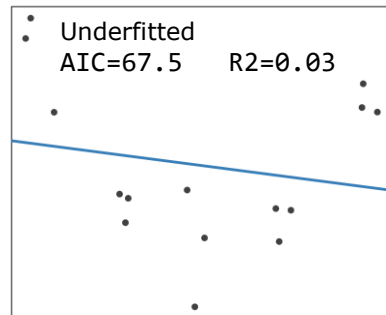
Frequentist AIC

„Akaike information criterion“ more flexible than F-tests

$$\text{AIC} = -2 \cdot \log(L) + 2 \cdot k$$

- Computed from likelihood L (remember: *maximum* likelihood)
Model with higher likelihood-value fits the data better
- Adds a penalty term for model complexity k (number of parameters)
→ Model with lower AIC is better

„Principle of parsimony“



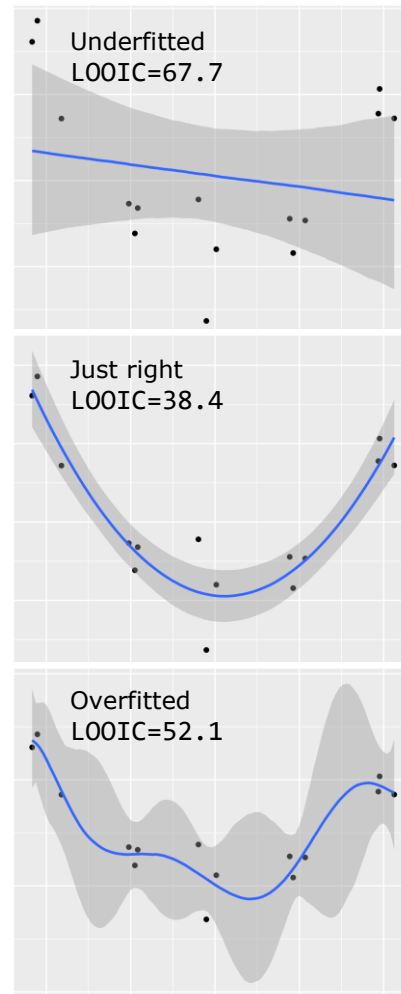
Bayesian LOO

“Leave-one-out cross-validation”

elpd = expected log predictive density

- Computed from likelihood & posterior
- Includes parameter uncertainty & penalizes model complexity
- Estimates for how well the model would predict for a new dataset
→ Model with higher elpd = better

$\text{LOOIC} = -2 \cdot \text{elpd}$ (lower values = better) just for convenience
for people used to AIC



Bayesian LOO

```
fit2 = brm(log(height)~z.lat+z.rain)
```

```
fit3 = brm(log(height)~z.lat*z.rain)
```

```
> LOO(fit3)
```

Computed from 4000 by 131 log-likelihood matrix.

	Estimate	SE	
elpd_loo	-235.9	6.9	elpd: larger values are better
p_loo	3.5	0.4	p: effective number of parameters
looic	471.8	13.9	looic=-2*elpd

MCSE of elpd_loo is 0.0.

MCSE and ESS estimates assume MCMC draws (r_{eff} in $[0.7, 1.2]$).

All Pareto k estimates are good ($k < 0.7$).

See `help('pareto-k-diagnostic')` for details.

Estimates come with standard error

```
> LOO(fit2, fit3)
```

Model comparisons:

	elpd_diff	se_diff	
fit3	0.0	0.0	Best elpd shown on top
fit2	-3.5	2.4	

Difference in elpd is associated with uncertainty

When $\text{elpd_diff} > 2 * \text{se_diff}$ (approximately),
you can be sure the model is better.

Here, both models perform equally under uncertainty,
so we would choose the less complex one (fit2)

→ Use model comparison with LOO (similar to AIC) in the Bayesian framework

1 Categorical predictor

1 predictor with 2 levels

Example: bird species richness vs. landscape type

Observed bird species richness in different habitats

Each habitat categorized by landscape type:

- Agriculture
- Urban
- Bauxite
- Forest

Start with subset Agriculture / Urban first

Later, we also include area as a predictor



1 predictor with 2 levels

Example: bird species richness vs. landscape type

Stochastic part: $S \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = \mu(\text{landscape})$

Each datapoint is a landscape patch

Categorical predictor: *landscape* (2 levels)

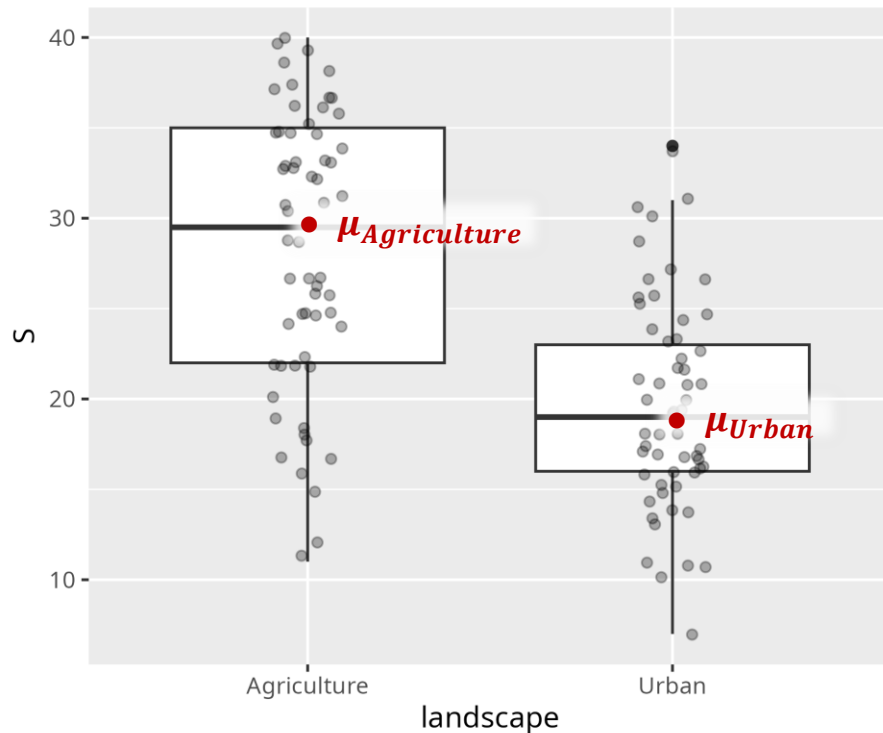
3 parameters: $\mu_{\text{Agriculture}}, \mu_{\text{Urban}}, \sigma$

Estimate and compare group-level means

Frequentist method: t-test

Does not compare distributions (overlap).

Compares their **means** !



Dummy coding

Deterministic part: $\mu = b_0 + b_1 \cdot x_{Urban}$

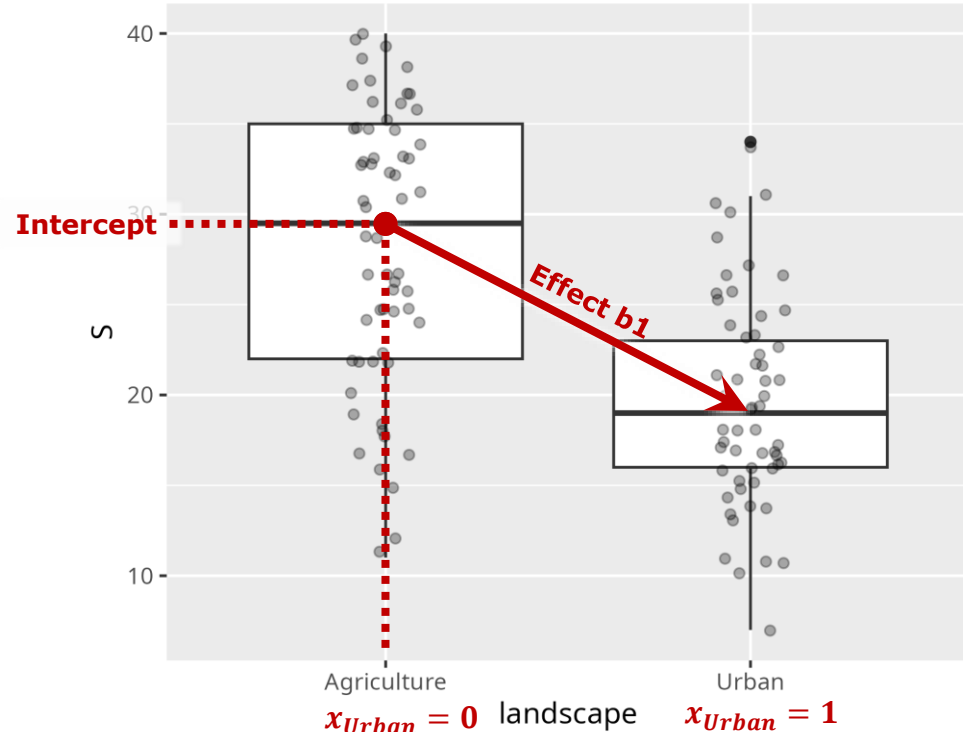
landscape = *Agriculture* is reference level

$$x_{Urban} = \begin{cases} 1 & \text{landscape} = \text{Urban} \\ 0 & \text{otherwise} \end{cases}$$

$$\mu_{Agriculture} = b_0 + b_1 \cdot 0 = b_0$$

$$\mu_{Urban} = b_0 + b_1 \cdot 1 = b_0 + b_1$$

→ Linear model with „intercept“ b_0 & „effect“ b_1



Model fitting

brms uses **dummy-coding** (effect-coding) as default

```
> brm( S ~ landscape )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	28.37	0.88	26.65	30.07
landscapeUrban	-8.77	1.31	-11.28	-6.14

Q: Is there a difference in means?

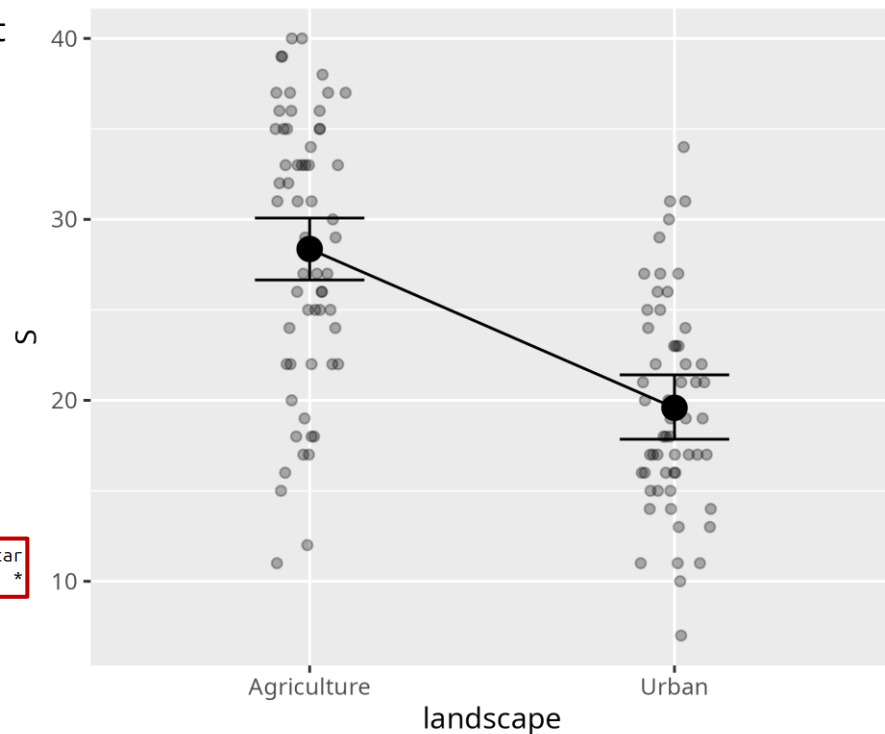
→ Look at posterior distribution of effect

```
> hypothesis(fit_a1, "landscapeUrban<0")
```

Hypothesis Tests for class b:

Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
1 (landscapeUrban) < 0	-8.77	1.31	-10.89	-6.59	Inf	1	*

$$P(\mu_{Urban} < \mu_{Agriculture}) = 1$$



Model fitting

But we could enforce **mean-coding**

```
> brm( S ~ landscape-1 )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
landscapeAgriculture	28.38	0.88	26.60	30.14
landscapeUrban	19.61	0.90	17.86	21.42

Q: Is there a difference in means?

→ Look at posterior distribution of difference

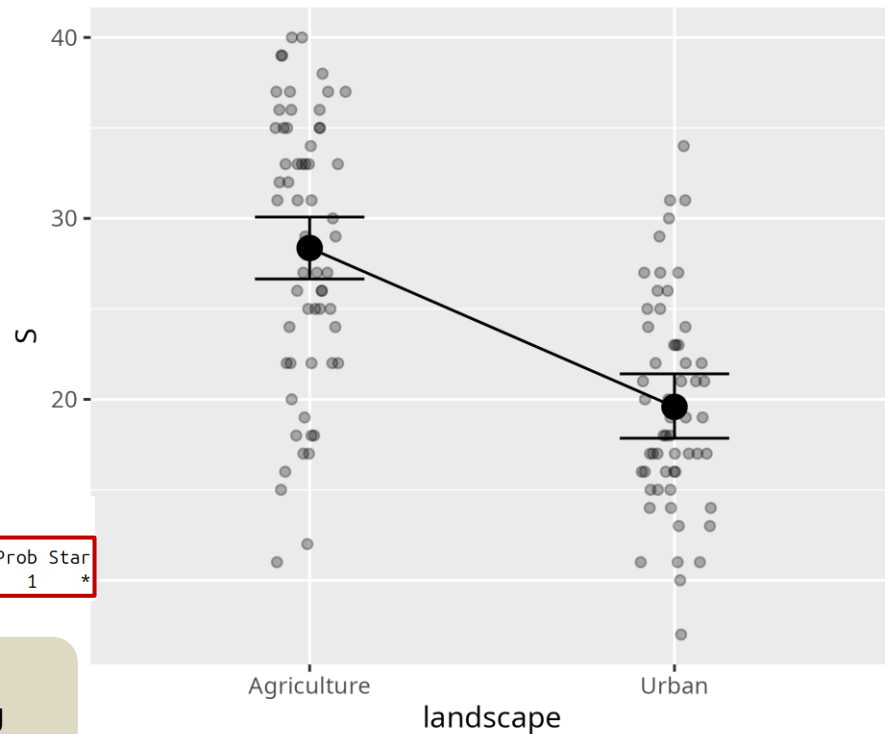
```
> hypothesis(fit_a1_1, "landscapeUrban<landscapeAgriculture")
```

Hypothesis Tests for class b:

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
1	(landscapeUrban)-... < 0	-8.77	1.26	-10.86	-6.72	Inf	1	*

$$P(\mu_{Urban} < \mu_{Agriculture}) = 1$$

dummy-coding
and mean-coding
are the same model !



1 predictor with K levels

Example: bird species richness vs. landscape type

Stochastic part: $S \sim \text{Normal}(\mu, \sigma)$

Deterministic part: $\mu = \mu(\text{landscape})$

Each datapoint is a landscape patch

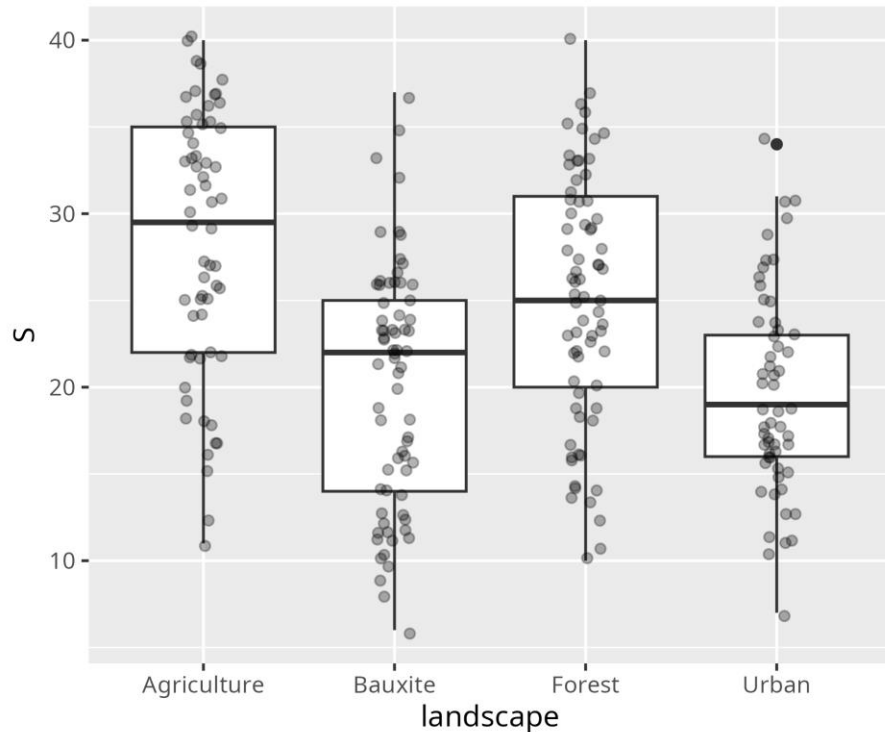
Categorical predictor: *landscape* (4 levels)

$K+1$ parameters: $\mu_{\text{Agriculture}}, \mu_{\text{Bauxite}},$
 $\mu_{\text{Forest}}, \mu_{\text{Urban}}, \sigma$

Estimate and compare group-level means

Frequentist method: F-test (ANOVA)

Test model against intercept-only model



Dummy coding with K levels

$$\mu = b_0 + b_1 \cdot x_{Bauxite} + b_2 \cdot x_{Forest} + b_3 \cdot x_{Urban}$$

landscape = *Agriculture* is reference level

$K-1$ dummy variables:

$$x_{Bauxite} = \begin{cases} 1 & \text{landscape} = \text{Bauxite} \\ 0 & \text{otherwise} \end{cases}$$

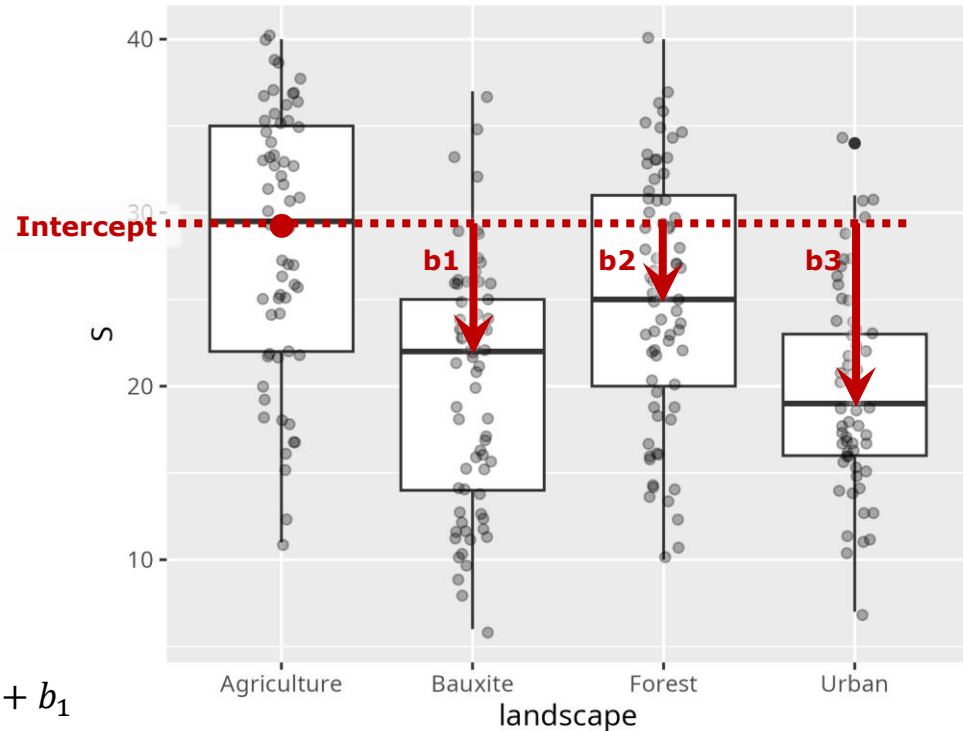
$$x_{Forest} = \begin{cases} 1 & \text{landscape} = \text{Forest} \\ 0 & \text{otherwise} \end{cases}$$

$$x_{Urban} = \begin{cases} 1 & \text{landscape} = \text{Urban} \\ 0 & \text{otherwise} \end{cases}$$

$$\mu_{Agriculture} = b_0 + b_1 \cdot 0 + b_2 \cdot 0 + b_3 \cdot 0 = b_0$$

$$\mu_{Bauxite} = b_0 + b_1 \cdot 1 + b_2 \cdot 0 + b_3 \cdot 0 = b_0 + b_1$$

etc ...



Model fitting with K levels

```
> brm( S ~ landscape )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	28.38	0.88	26.58	30.05
landscapeBauxite	-8.33	1.22	-10.63	-5.84
landscapeForest	-3.35	1.21	-5.67	-0.97
landscapeUrban	-8.78	1.29	-11.31	-6.25

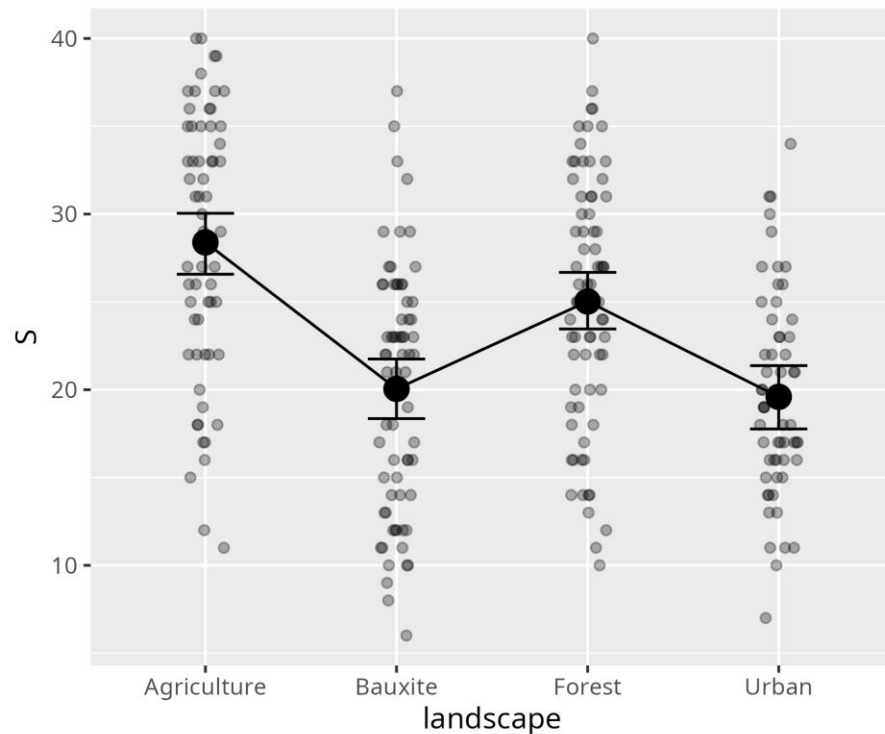
Q: Is there a difference in means?

- Individual effects don't give an overall answer
- Compare against intercept-only model (similar to frequentist F-test)

```
> LOO(fit_landscape, fit_intercept)
```

Model comparisons:

	elpd_diff	se_diff	
fit_landscape	0.0	0.0	Yes, $S \sim \text{landscape}$ is a
fit_intercept	-27.1	7.4	better model than $S \sim 1$



2 Categorical predictors

2 predictors with K & L levels

Example: S vs. landscape type & area size

Additive model $S \sim \text{landscape} + \text{area}$

Accounts for difference in area size

Area effect is the same for all landscape levels

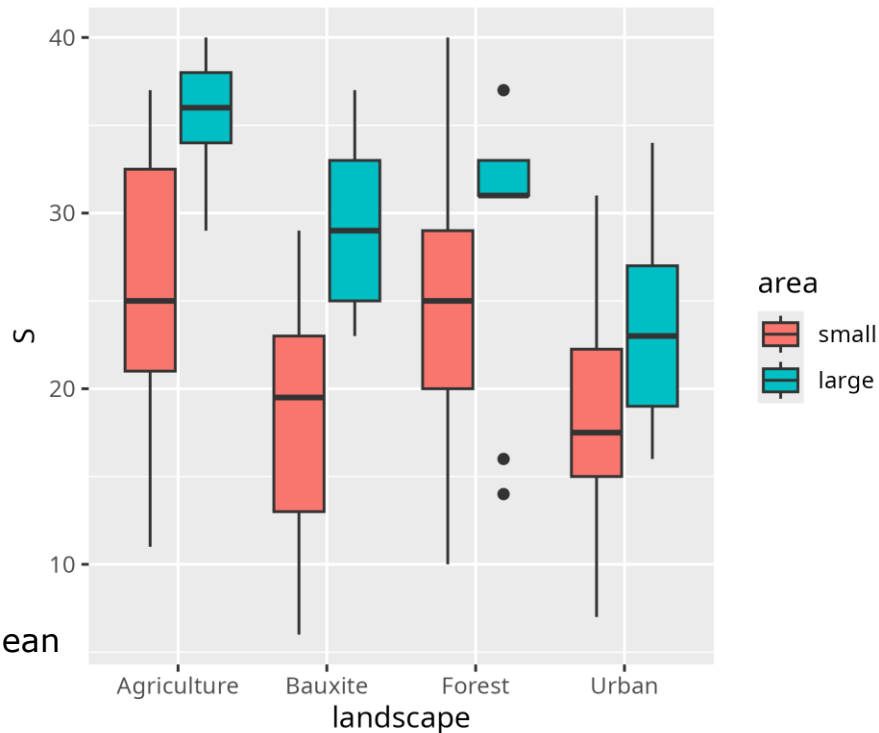
$K + L - 1$ parameters (+1 for sdev)

Factorial model $S \sim \text{landscape} * \text{area}$

Area effect changes over landscape levels

Each landscape:size combination is fitted with own mean

$K \cdot L$ parameters (+1 for sdev)



Additive model

$$\mu = b_0 + b_1 \cdot x_{\text{Bauxite}} + b_2 \cdot x_{\text{Forest}} + b_3 \cdot x_{\text{Urban}} + \textcolor{red}{b_4 \cdot x_{\text{large}}}$$

landscape = Agriculture, area = small is reference level

1 intercept

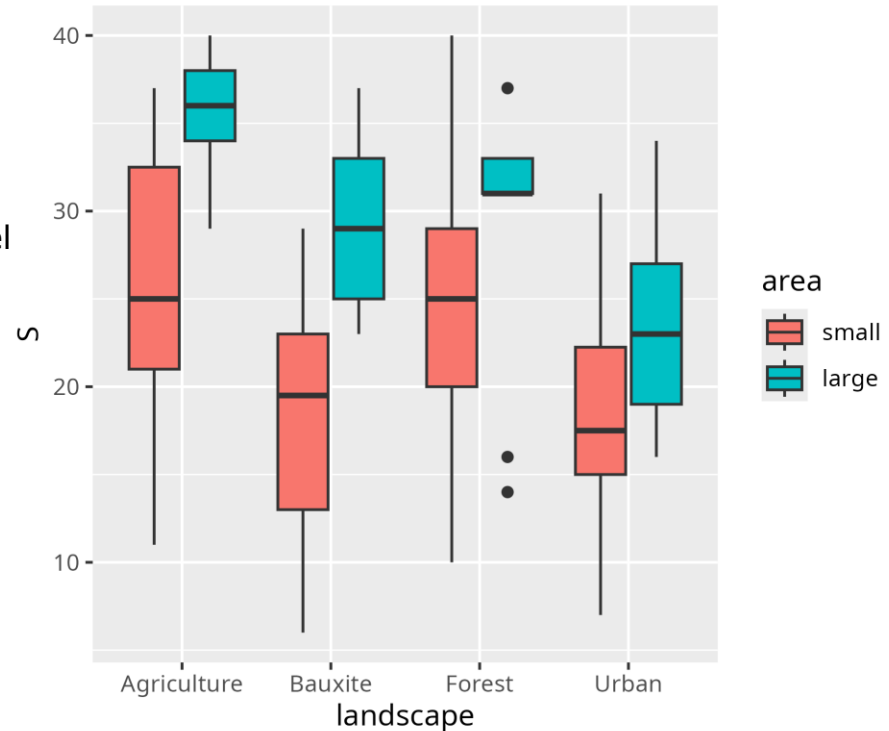
$K-1$ dummy variables for landscape

$L-1$ dummy variables for area

$=K+L-1$ variables

less than $K*L$ level combinations

→ Will not fit independent group-level means



Additive model

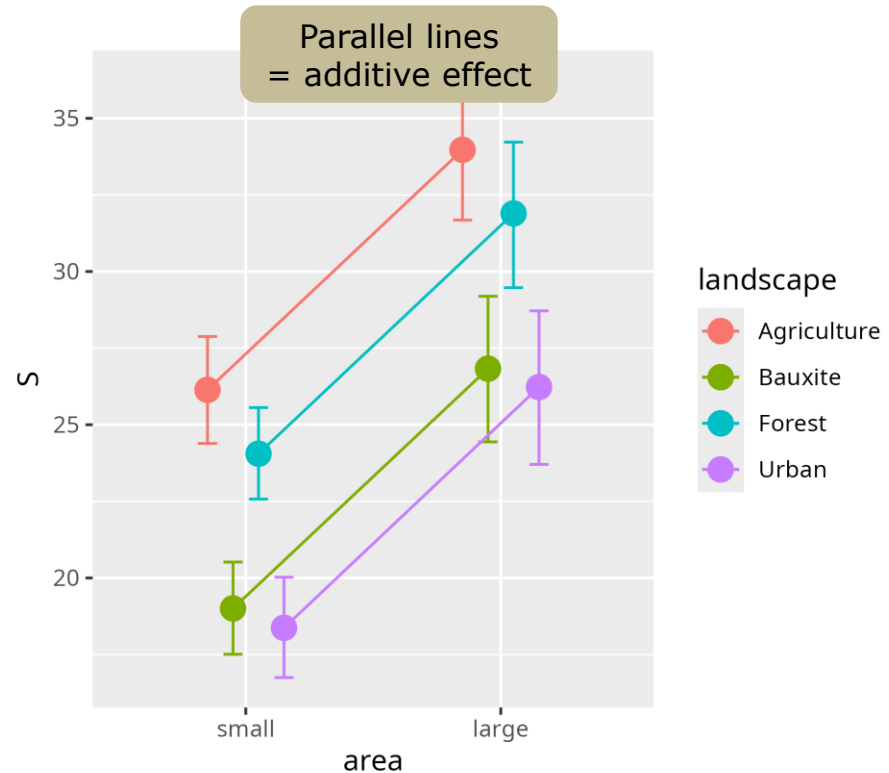
```
> brm( S ~ landscape + area )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	26.14	0.89	24.39	27.88
landscapeBauxite	-7.13	1.15	-9.36	-4.84
landscapeForest	-2.07	1.14	-4.25	0.16
landscapeUrban	-7.75	1.20	-10.10	-5.46
arealarge	7.82	1.10	5.62	9.98

Q: Is there an additional effect of patch size?

→ Strong effect of area
7.82 more species in large patches (on avg.)



Additive model

```
> brm( S ~ landscape + area )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	26.44	0.88	24.38	27.88
landscapeBauxite	-7.11	1.10	-9.29	-4.84
landscapeForest	-2.11	1.10	-4.29	-0.16
landscapeUrban	-7.11	1.10	-9.29	-4.84
arealarge	7.82	1.10	5.62	9.98

Summary table not
helpful for predictors
with >2 levels

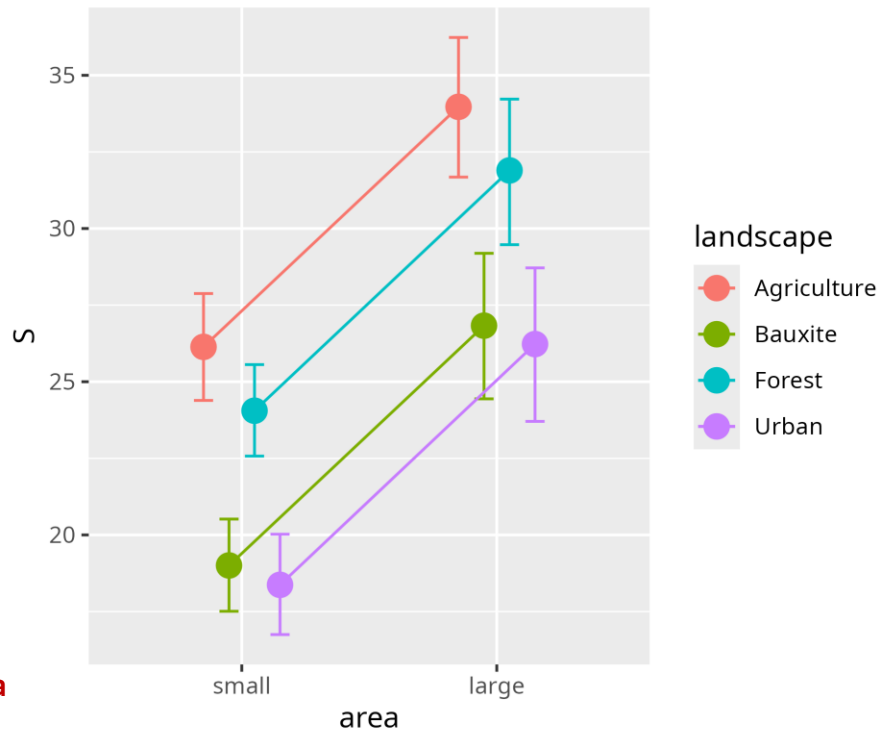
Q: Is there an additional effect of patch size?

→ Model comparison

```
> LOO(fit_additive, fit_landscape)
```

Model comparisons:

	elpd_diff	se_diff	Yes, $S \sim \text{landscape} + \text{area}$ is a better model than $S \sim \text{landscape}$
fit_additive	0.0	0.0	
fit_landscape	-23.7	6.9	



Additive model

```
> brm( S ~ landscape + area )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	26.44	0.88	24.38	27.88
landscapeBauxite	-7.10	1.10	-9.29	-4.84
landscapeForest	-2.16	1.16	-4.46	-0.46
landscapeUrban	-7.46	1.46	-9.98	-4.98
arealarge	7.82	1.10	5.62	9.98

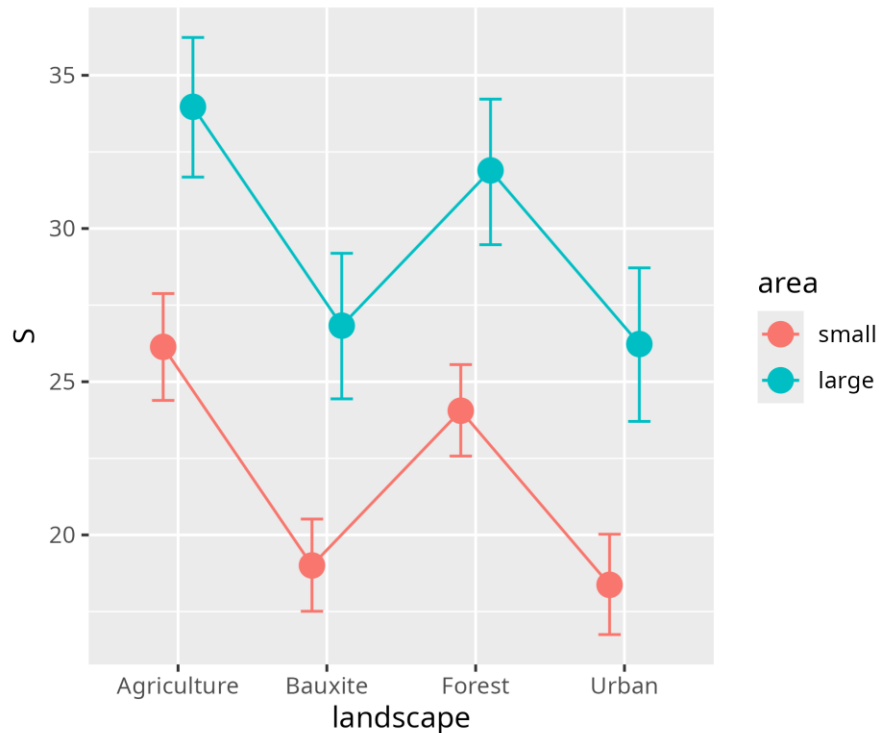
Summary table not
helpful for predictors
with >2 levels

Q: Are there differences between landscapes,
when controlling for area size?

```
> LOO(fit_additive, fit_area)
```

Model comparisons:

	elpd_diff	se_diff	Yes, $S \sim \text{landscape} + \text{area}$ is a better model than $S \sim \text{area}$
fit_additive	0.0	0.0	
fit_area	-26.9	7.1	



Factorial model

$$\begin{aligned}\mu = & b_0 + \\ & b_1 \cdot x_{\text{Bauxite}} + b_2 \cdot x_{\text{Forest}} + b_3 \cdot x_{\text{Urban}} + \\ & b_4 \cdot x_{\text{large}} + \\ & \textcolor{red}{b_5 \cdot x_{\text{Bauxite,large}} + b_6 \cdot x_{\text{Forest,large}} + b_7 \cdot x_{\text{Urban,large}}}\end{aligned}$$

landscape = Agriculture, area = small is reference level

1 intercept

$K-1$ dummy variables for landscape

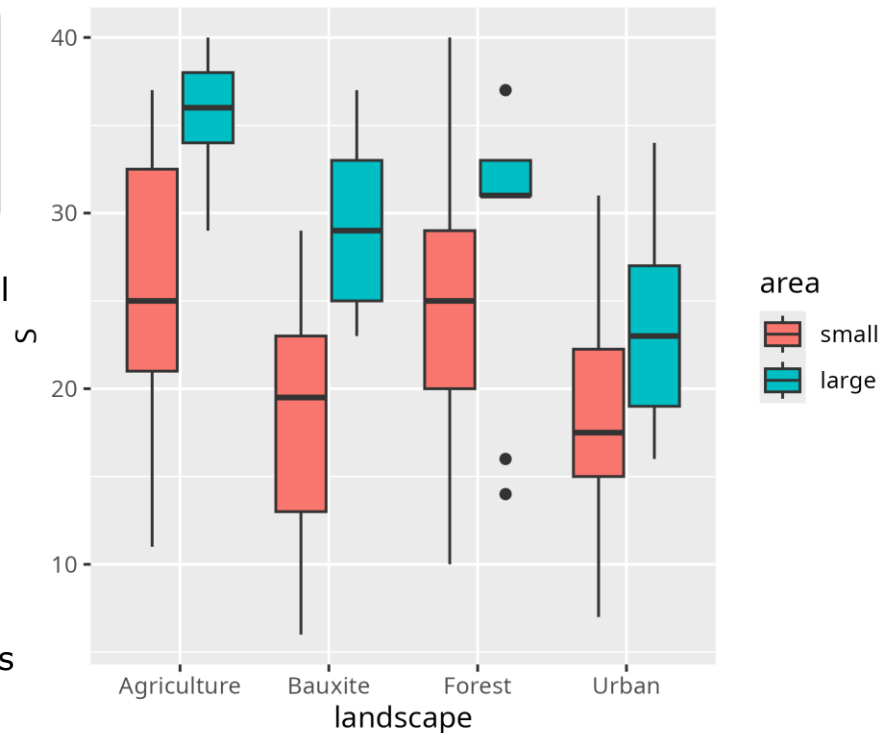
$L-1$ dummy variables for area

$(K-1) \cdot (L-1)$ dummy variables for landscape:area

= $K \cdot L$ variables in total

→ Fitting independent means to all level combinations

$$\mu = \mu(\text{landscape}, \text{area})$$



Factorial model

```
> brm( S ~ landscape * area )
```

Regression Coefficients:

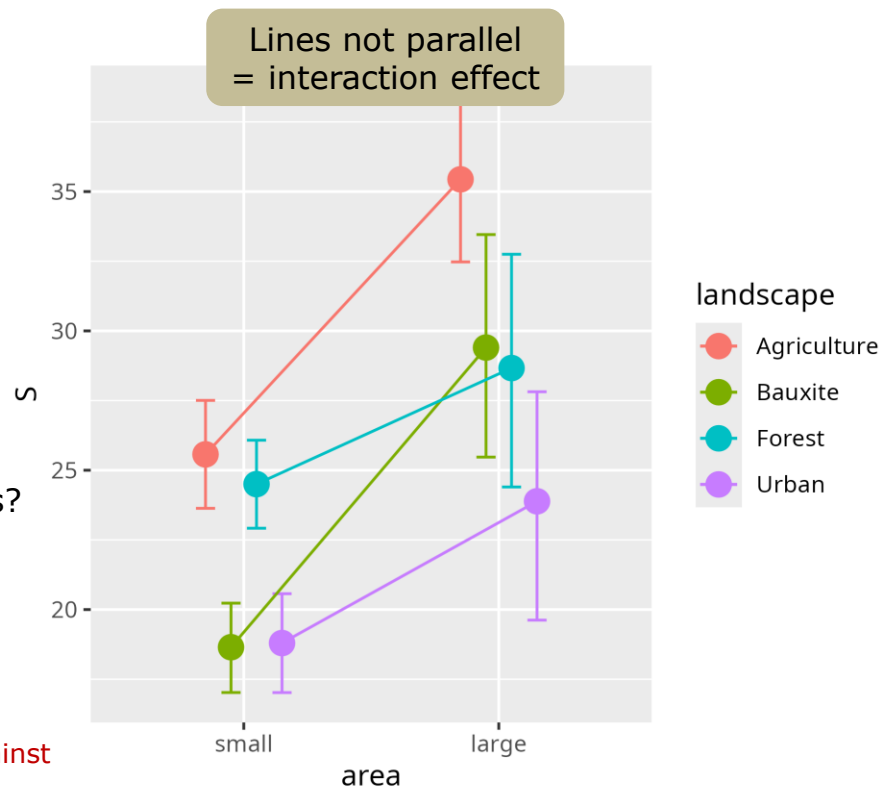
	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	25.56	0.97	23.63	27.51
landscapeBauxite	-6.93	1.27	-9.43	-4.47
landscapeForest	-1.06	1.24	-3.50	1.37
landscapeUrban	-6.78	1.35	-9.44	-4.08
arealarge	9.89	1.81	6.32	13.47
landscapeBauxite:arealarge	0.87	2.87	-4.56	6.46
landscapeForest:arealarge	-5.76	2.91	-11.69	-0.29
landscapeUrban:arealarge	-4.81	2.91	-10.52	0.97

Q: Does area effect change between landscape levels?

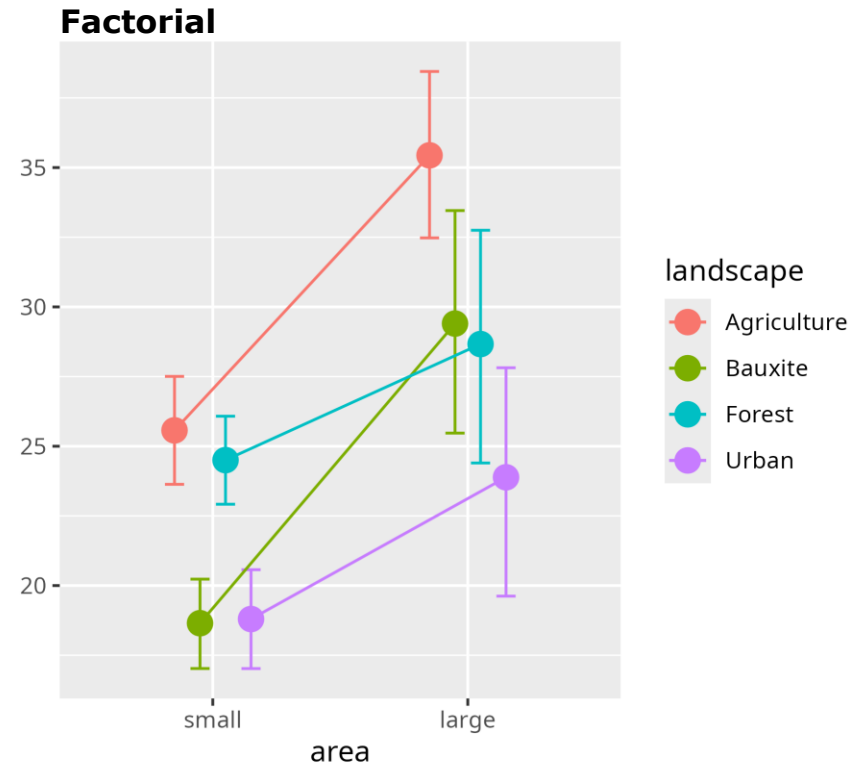
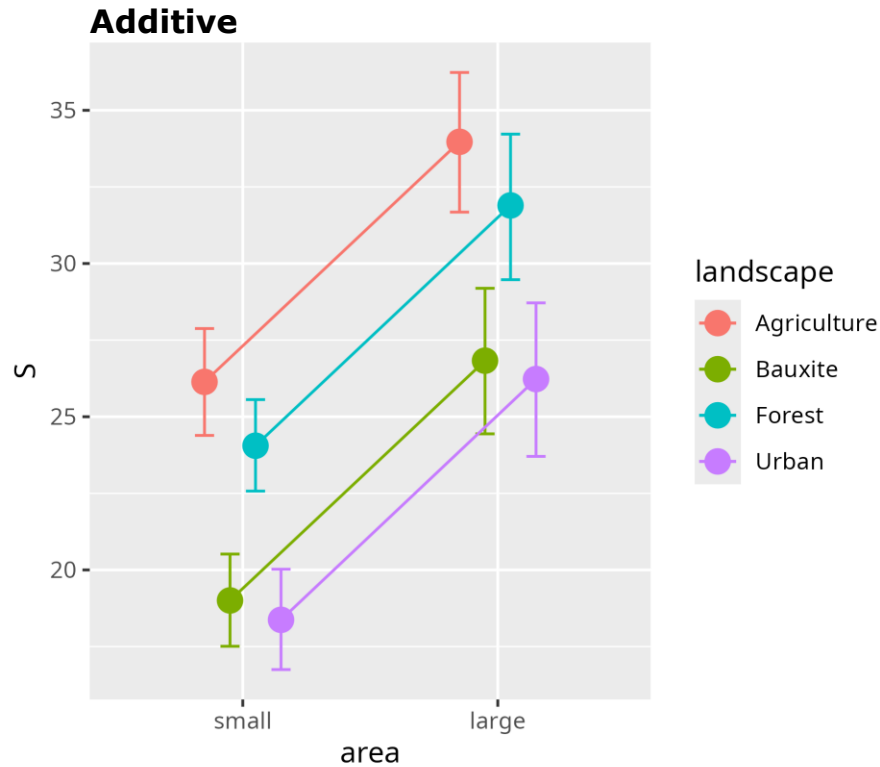
```
> LOO(fit_factorial, fit_additive)
```

Model comparisons:

	elpd_diff	se_diff	
fit_factorial	0.0	0.0	No strong evidence for
fit_additive	-0.7	2.5	S~landscape*area against S~landscape+area



Factorial vs additive



No strong evidence for interaction found → select additive as the best model

Post-hoc analysis

What is post-hoc analysis?

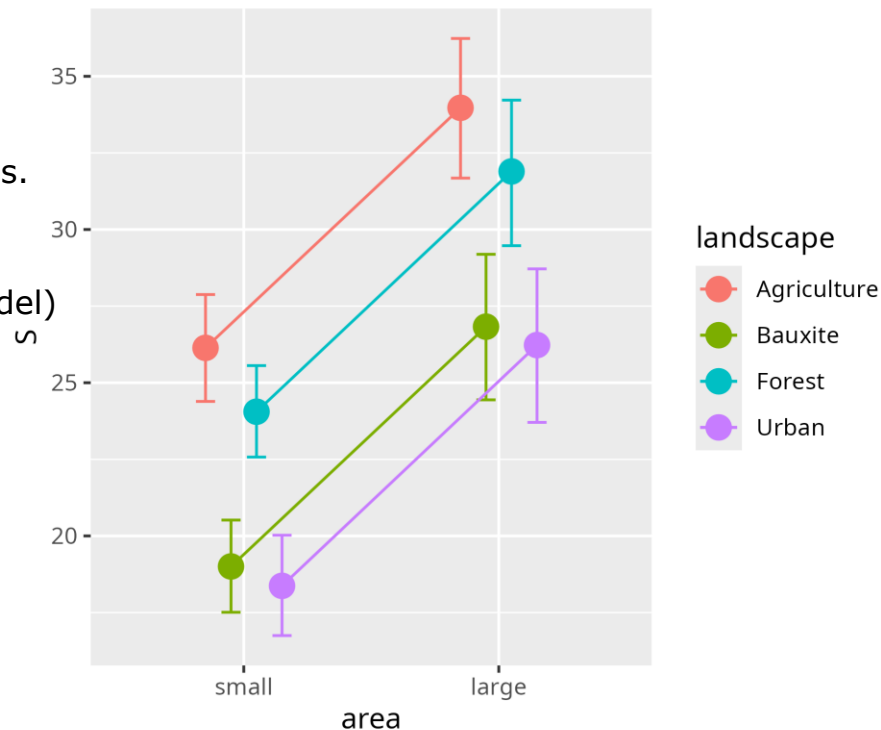
Model comparison (LOO)

tells you **IF** there is a difference between group-levels.

Post-hoc analysis (**after** selecting an appropriate model)
tells you **WHAT** the difference is.

Analysis is **model-based**.

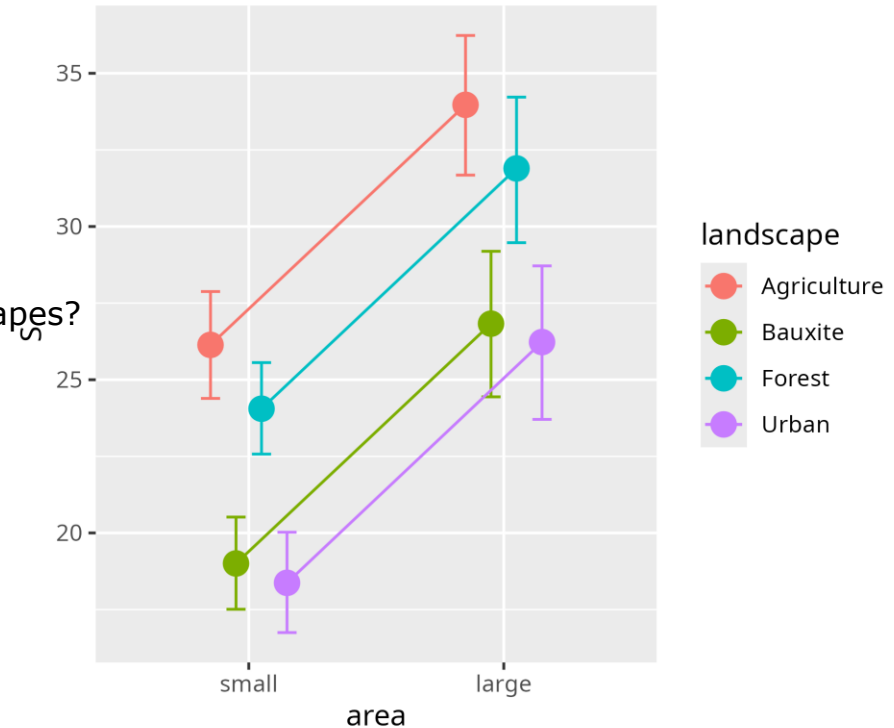
Do not just compute empirical means from the data.



What is post-hoc analysis?

Answer questions like:

- What is the mean species richness in small areas?
→ Average over landscapes.
- What is the mean species richness in urban landscapes?
→ Average over area sizes.
- What is the mean difference between urban and agricultural landscapes?
- And what are all their associated uncertainties?



Bayesian post-hoc analysis

Make predictions & compute their average or difference etc depending on the question.

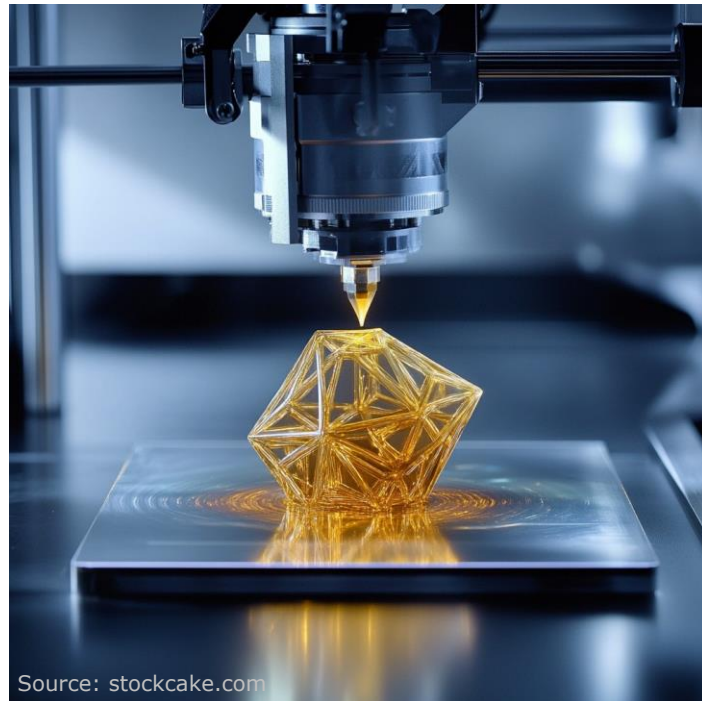
Remember: everything is a distribution!

- For each sample k of the posterior ($k = 1 \dots 1000$)
use it's predictions, compute what's required, e.g. $a_k - b_k$
- That is a sample of posterior distribution for $a - b$
- Compute mean, standard deviation, quantiles, etc

→ The **emmeans** package can automate these steps!

→ Alternative: **marginalEffects** package.
Powerful but a bit more complex

The Bayesian 3D printer



Source: stockcake.com

Post-hoc analysis: 1 predictor

```
> brm( S ~ landscape )
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	28.38	0.88	26.58	30.05
landscapeBauxite	-8.33	1.22	-10.63	-5.84
landscapeForest	-3.35	1.21	-5.67	-0.97
landscapeUrban	-8.78	1.29	-11.31	-6.25

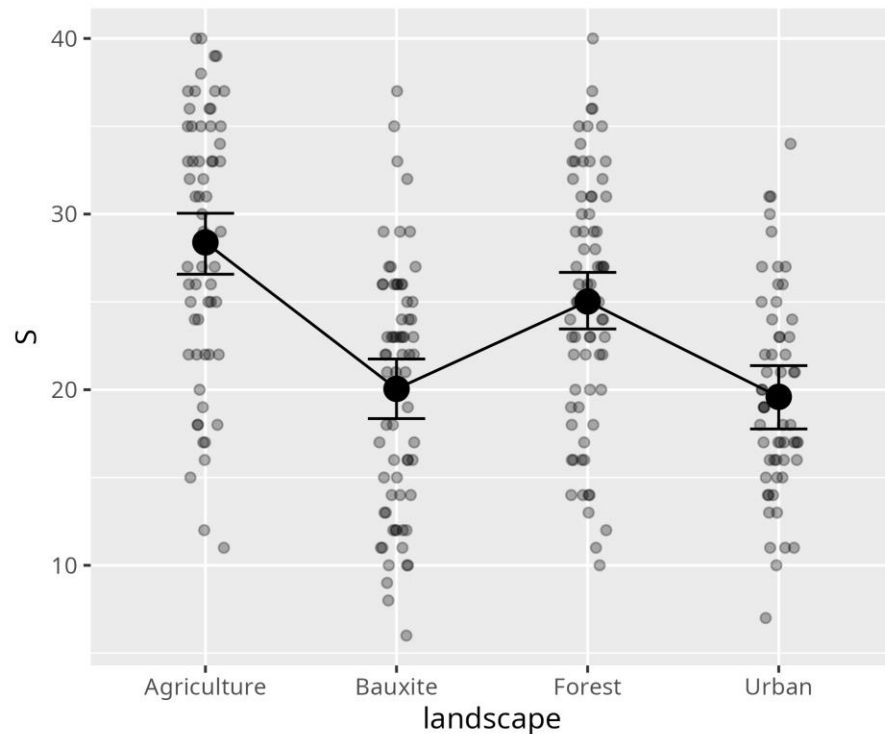
Q: What are the predicted means
(and their uncertainties)?

```
> emmeans(fit_landscape, "landscape")
```

landscape	emmean	lower.HPD	upper.HPD
Agriculture	28.4	26.6	30.0
Bauxite	20.0	18.5	21.8
Forest	25.0	23.4	26.6
Urban	19.6	17.8	21.4

Point estimate displayed: median

HPD interval probability: 0.95



Post-hoc analysis: 1 predictor

```
> brm( S ~ landscape )
```

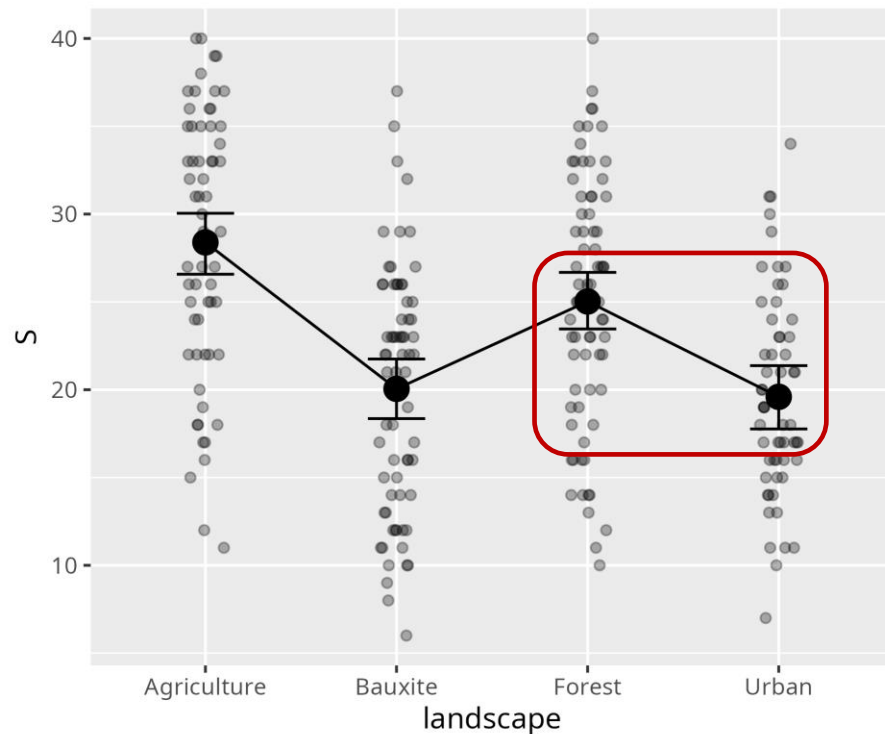
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	28.38	0.88	26.58	30.05
landscapeBauxite	-8.33	1.22	-10.63	-5.84
landscapeForest	-3.35	1.21	-5.67	-0.97
landscapeUrban	-8.78	1.29	-11.31	-6.25

Q: What is the difference between forest & urban?

```
> emmeans(fit_landscape, "landscape") |> pairs()
```

contrast	estimate	lower.HPD	upper.HPD
Agriculture - Bauxite	8.35	5.857	10.56
Agriculture - Forest	3.30	0.966	5.63
Agriculture - Urban	8.73	6.290	11.24
Bauxite - Forest	-5.00	-7.217	-2.60
Bauxite - Urban	0.45	-2.232	2.86
Forest - Urban	5.45	2.940	7.83



Point estimate displayed: median
HPD interval probability: 0.95

Post-hoc analysis: multiple predictors

```
> brm( S ~ landscape + area )
```

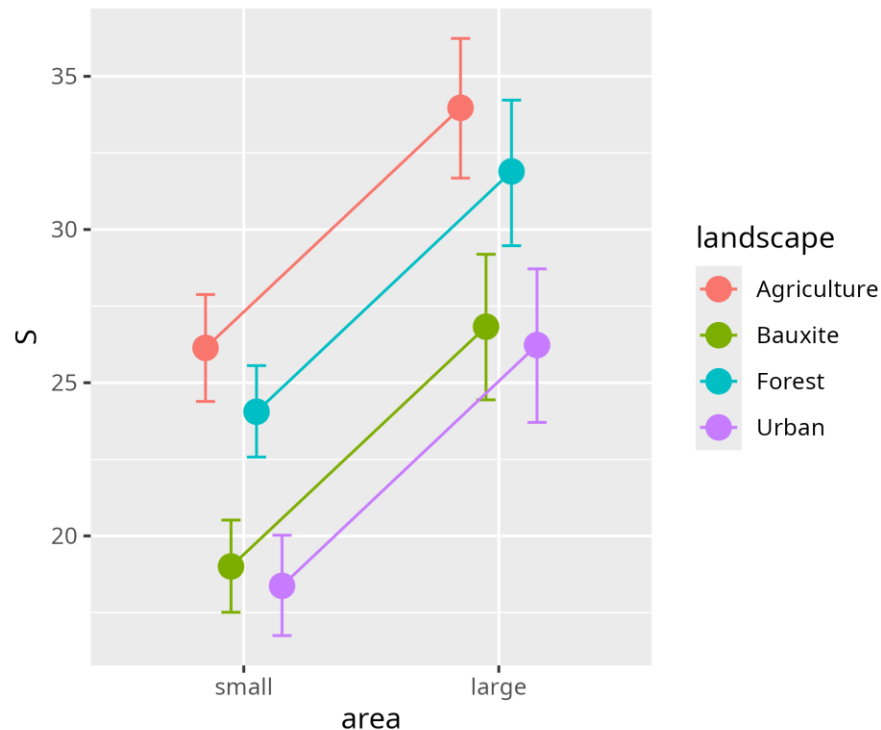
Q: What are group-level means?

```
> emmeans(fit_additive, ~area*landscape)
```

area	landscape	emmean	lower.HPD	upper.HPD
small	Agriculture	26.2	24.4	27.9
large	Agriculture	33.9	31.7	36.1
small	Bauxite	19.0	17.6	20.6
large	Bauxite	26.8	24.6	29.2
small	Forest	24.1	22.6	25.6
large	Forest	31.9	29.5	34.3
small	Urban	18.4	16.8	20.2
large	Urban	26.2	23.8	28.5

Point estimate displayed: median

HPD interval probability: 0.95



Post-hoc analysis: multiple predictors

```
> brm( S ~ landscape + area )
```

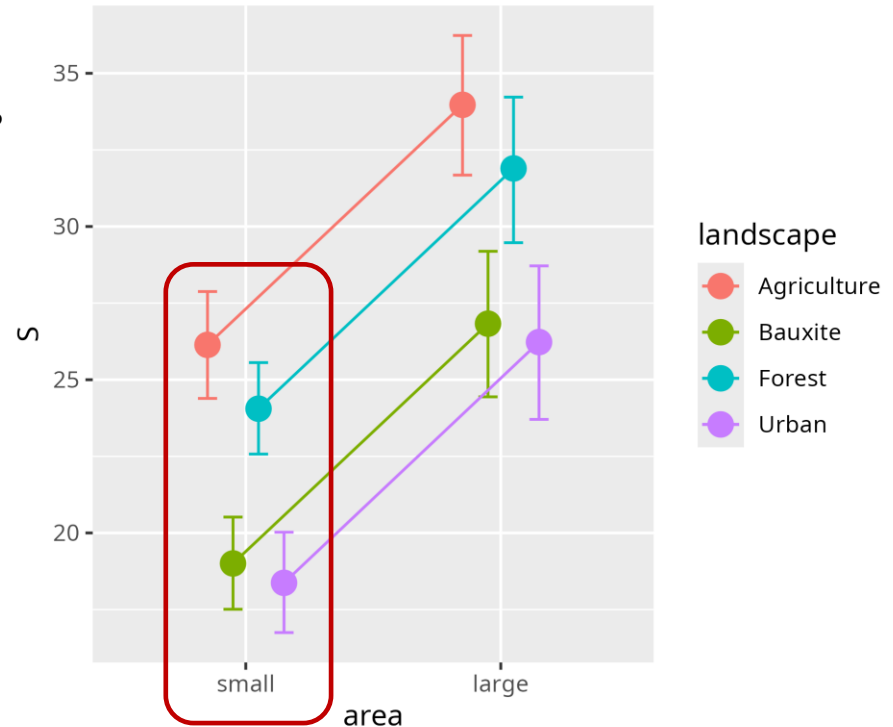
Q: What is the mean species richness in small areas?

```
> emmeans(fit_additive, ~area)
area  emmean lower.HPD upper.HPD
small   21.9    21.0    22.7
large   29.7    27.8    31.6
```

Results are averaged over the levels of: landscape

Point estimate displayed: median

HPD interval probability: 0.95



Post-hoc analysis: multiple predictors

```
> brm( S ~ landscape + area )
```

Q: What is the mean species richness in urban landscapes?

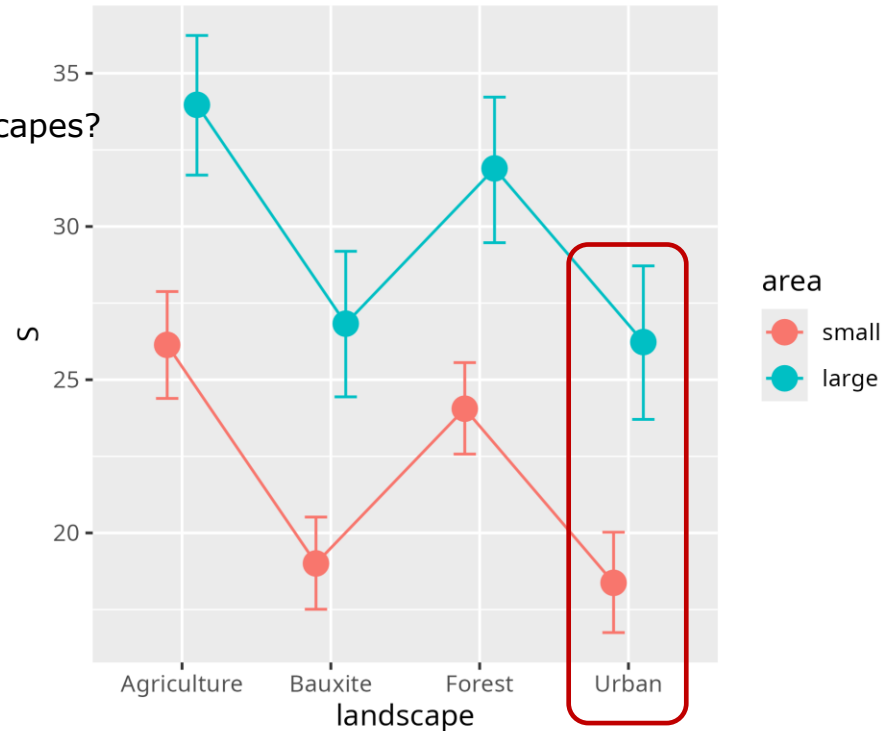
```
> emmeans(fit_additive, ~landscape)
```

landscape	emmean	lower.HPD	upper.HPD
Agriculture	30.0	28.4	31.7
Bauxite	22.9	21.3	24.6
Forest	28.0	26.3	29.7
Urban	22.3	20.5	24.2

Results are averaged over the levels of: area

Point estimate displayed: median

HPD interval probability: 0.95



Post-hoc analysis: multiple predictors

```
> brm( S ~ landscape + area )
```

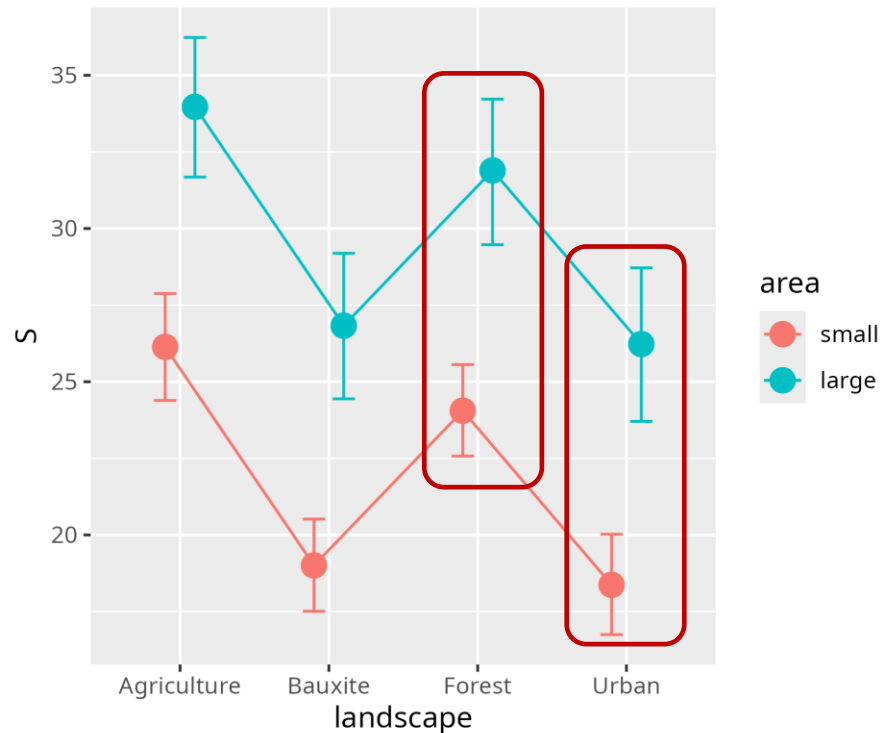
Q: What is the mean difference between urban and forest landscapes?

```
> emmeans(fit_additive, ~landscape) |> pairs()
contrast      estimate lower.HPD upper.HPD
Agriculture - Bauxite    7.128    4.9378    9.35
Agriculture - Forest     2.073   -0.0856    4.36
Agriculture - Urban     7.779    5.4187   10.13
Bauxite - Forest    -5.064   -7.1364   -2.98
Bauxite - Urban     0.653   -1.6066    2.70
Forest - Urban      5.721    3.4428    7.91
```

Results are averaged over the levels of: area

Point estimate displayed: median

HPD interval probability: 0.95



*Categorical & continuous predictors
(ANCOVA)*

Categorical & continuous predictor

Example: bird species richness

A lot of unexplained variation in $S \sim \text{landscape}$

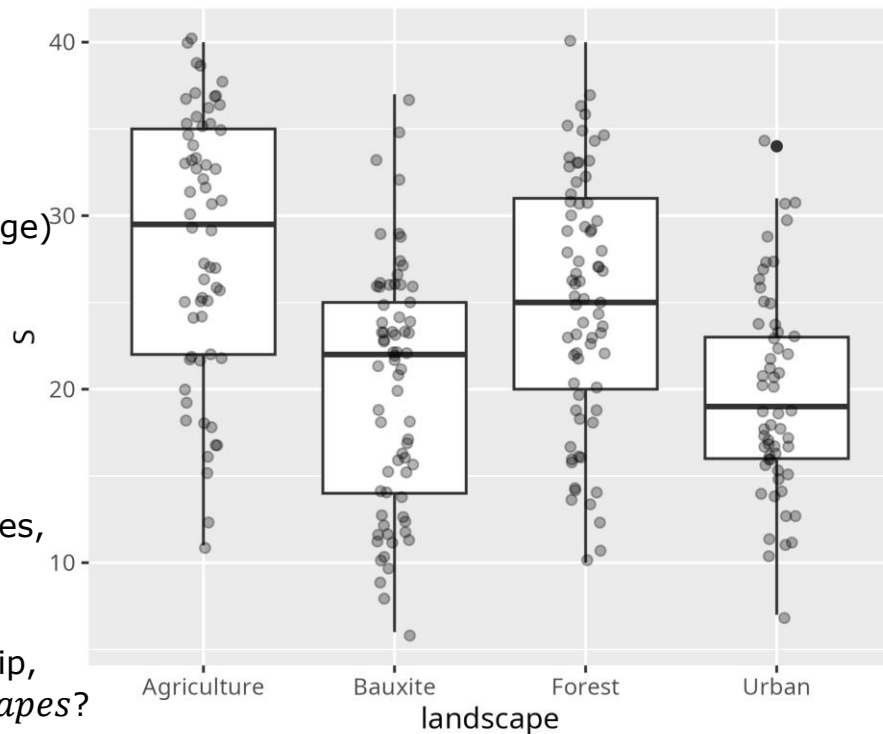
Last section: added predictor *area* (levels: small, large)
 $S \sim \text{landscape} + \text{area}$

If we have better resolved data for area (in km²)

→ Use **continuous predictor** *log.area*

Q: Is species richness S different over *landscape* types, while controlling for *log.area*?

Q: How strong is the average species-area relationship, while controlling (acknowledging) different *landscapes*?



Categorical & continuous predictor

Example: S vs. landscape type & log.area

Fit a regression line to each landscape level

Additive model $S \sim \text{landscape} + \log.\text{area}$

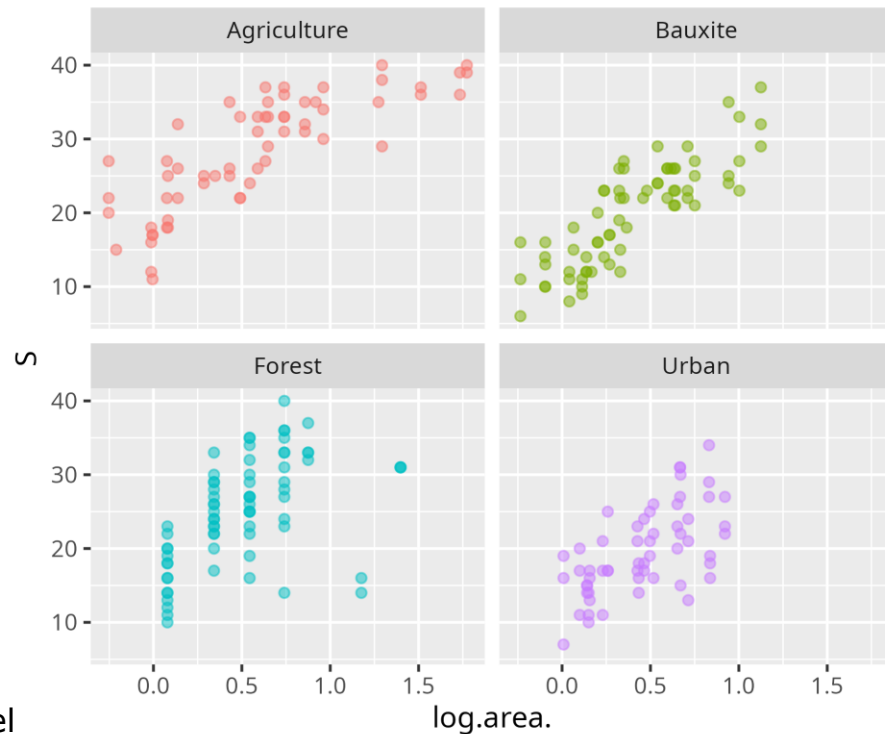
Slope (log.area) independent of landscape
→ identical slope

Individual intercepts for each landscape level

Factorial model $S \sim \text{landscape} * \log.\text{area}$

Slope (log.area) depends on landscape

Individual intercepts & slopes for each landscape level



Additive model

$S \sim \text{landscape} + \log.\text{area}$

$$\mu = \alpha(\text{landscape}) + \beta \cdot \log.\text{area}$$

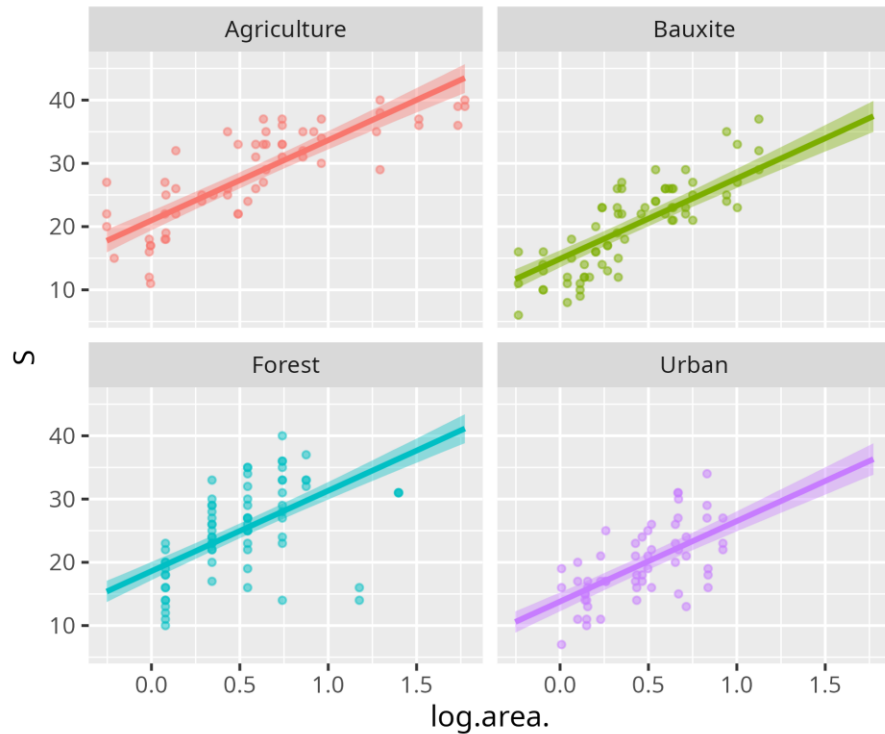
4 intercepts: $\alpha_{\text{Agriculture}}, \alpha_{\text{Bauxite}}, \alpha_{\text{Forest}}, \alpha_{\text{Urban}}$

1 slope: β

1 sdev: σ

Dummy-coding of intercepts:

$$\mu = a_0 + a_1 \cdot x_{\text{Bauxite}} + a_2 \cdot x_{\text{Forest}} + a_3 \cdot x_{\text{Urban}} + \beta \cdot \log.\text{area}$$



Additive model

```
> brm(S ~ landscape + log.area)
```

Regression Coefficients:

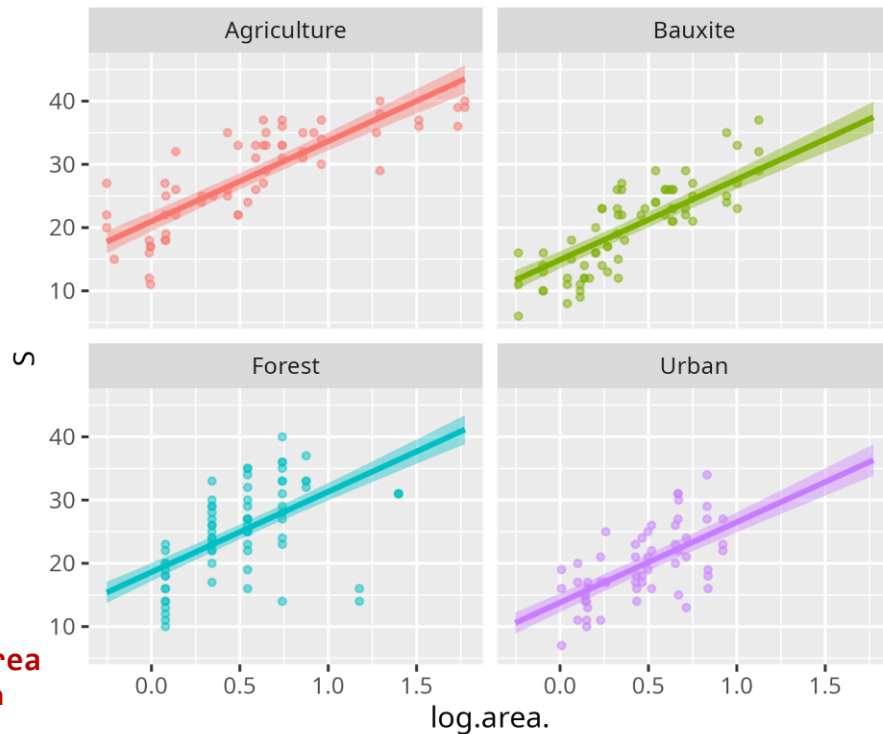
	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	20.96	0.79	19.42	22.48
landscapeBauxite	-6.06	0.90	-7.77	-4.23
landscapeForest	-2.34	0.90	-4.08	-0.56
landscapeUrban	-7.16	0.93	-9.00	-5.41
log.area.	12.71	0.81	11.09	14.29

Q: Is there a difference between landscape types, while accounting for area size?

```
L00(fit_additive, fit_logarea)
```

Model comparisons:

	elpd_diff	se_diff	Yes, $S \sim \text{landscape} + \log \text{area}$ is a better model than $S \sim \log \text{area}$
fit_additive	0.0	0.0	
fit_logarea	-31.8	7.7	



Additive model

```
> brm(S ~ landscape + log.area)
```

Regression Coefficients:

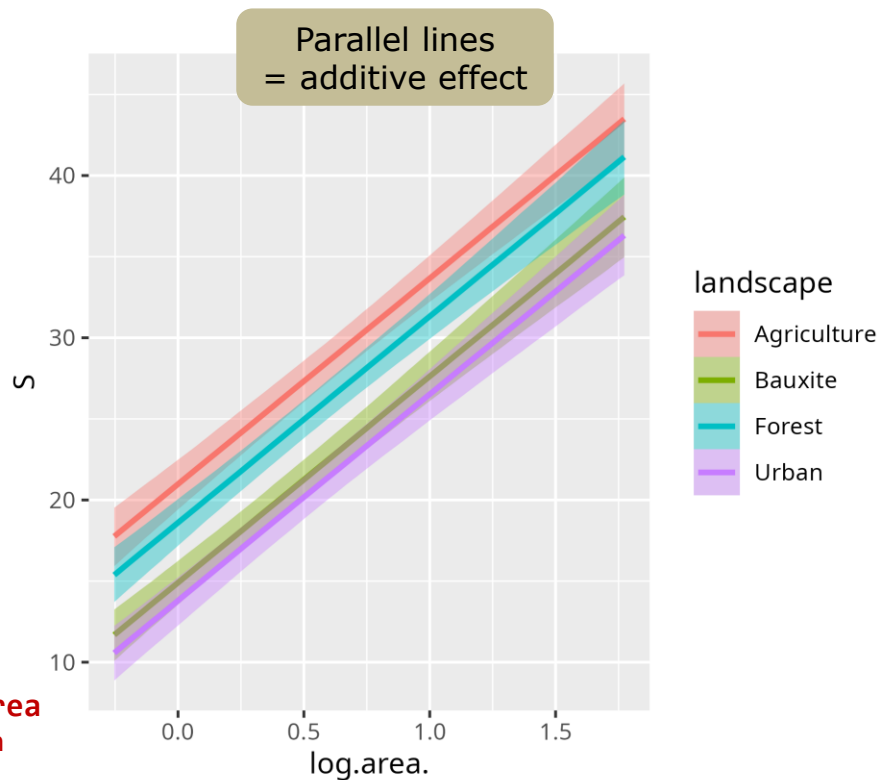
	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	20.96	0.79	19.42	22.48
landscapeBauxite	-6.06	0.90	-7.77	-4.23
landscapeForest	-2.34	0.90	-4.08	-0.56
landscapeUrban	-7.16	0.93	-9.00	-5.41
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	elpd_diff	se_diff	Yes, $S \sim \text{landscape} + \log \text{area}$ is a better model than $S \sim \log \text{area}$
fit_additive	0.0	0.0	
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Additive model

```
> brm(S ~ landscape + log.area)
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	20.96	0.79	19.42	22.48
landscapeBauxite	-6.06	0.90	-7.77	-4.23
landscapeForest	-2.34	0.90	-4.08	-0.56
landscapeUrban	-7.16	0.93	-9.00	-5.41
log.area.	12.71	0.81	11.09	14.29

Q: Is there a positive relation between S and $\log.area$, while acknowledging different *landscapes*?

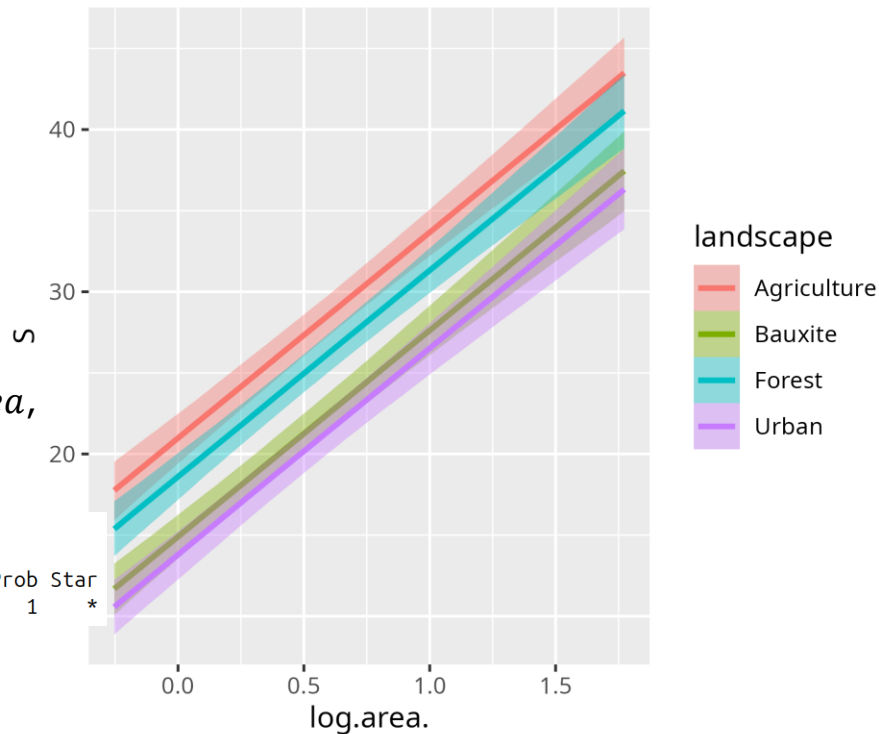
```
> hypothesis(fit_additive, "log.area.>0")
```

Hypothesis Tests for class b:

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
1	(log.area.) > 0	12.71	0.81	11.41	14.03	Inf	1	*

Yes, posterior distribution of slope positive

Alternatively, you could also do a model comparison (LOO)



Additive model

Post-hoc analysis

```
> brm(S ~ landscape + log.area)
```

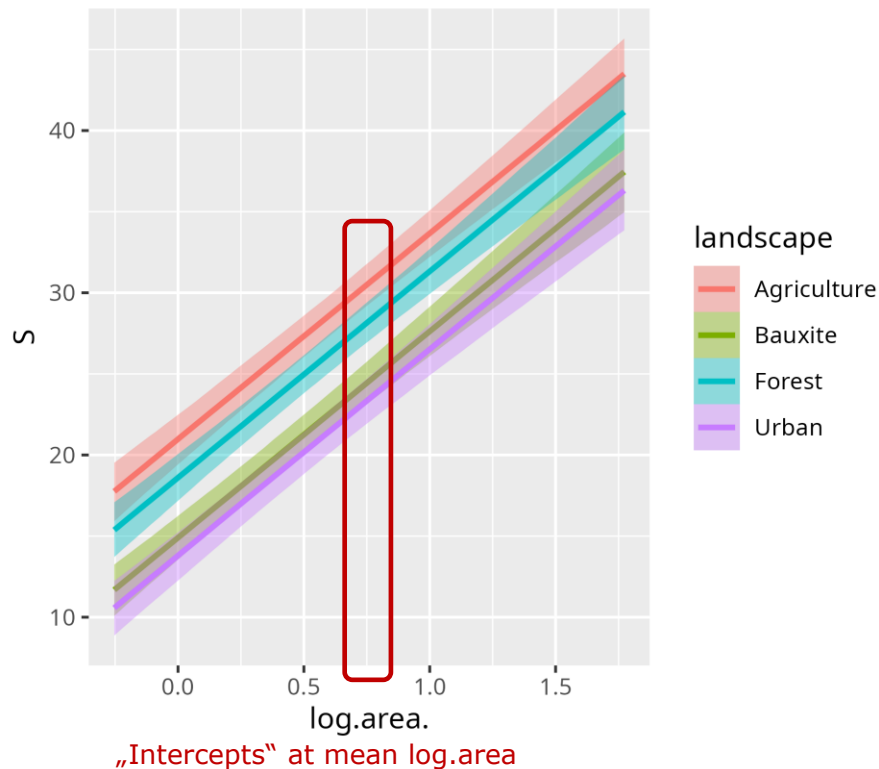
Q: What are mean intercepts and pairwise differences between landscape types?

```
> emmeans(fit_additive, ~landscape)
landscape   emmean lower.HPD upper.HPD
Agriculture 27.1    25.8    28.4
Bauxite     21.1    19.9    22.4
Forest      24.8    23.6    25.9
Urban       20.0    18.7    21.3
```

Point estimate displayed: median

HPD interval probability: 0.95

```
> emmeans(fit_additive, ~landscape) |> pairs()
contrast               estimate lower.HPD upper.HPD
Agriculture - Bauxite    6.05    4.301    7.80
Agriculture - Forest    2.35    0.648    4.13
Agriculture - Urban     7.14    5.426    9.02
Bauxite - Forest       -3.71   -5.312   -1.99
Bauxite - Urban        1.10   -0.557    3.01
Forest - Urban         4.78    3.104    6.62
```



Interaction model

$S \sim \text{landscape} * \text{log.area}$

$$\mu = \alpha(\text{landscape}) + \beta(\text{landscape}) \cdot \text{log.area}$$

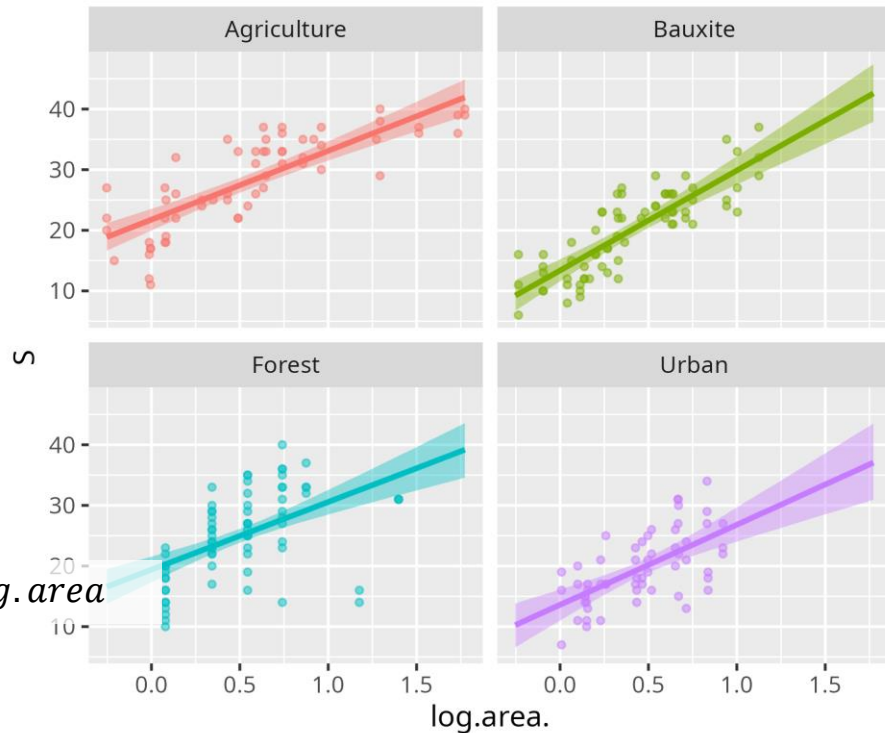
4 intercepts: $\alpha_{\text{Agriculture}}, \alpha_{\text{Bauxite}}, \alpha_{\text{Forest}}, \alpha_{\text{Urban}}$

4 slopes: $\beta_{\text{Agriculture}}, \beta_{\text{Bauxite}}, \beta_{\text{Forest}}, \beta_{\text{Urban}}$

1 sdev: σ

Dummy-coding of intercepts & slopes:

$$\mu = a_0 + a_1 \cdot x_{\text{Bauxite}} + a_2 \cdot x_{\text{Forest}} + a_3 \cdot x_{\text{Urban}} + (b_0 + b_1 \cdot x_{\text{Bauxite}} + b_2 \cdot x_{\text{Forest}} + b_3 \cdot x_{\text{Urban}}) \cdot \text{log.area}$$



Interaction model

```
> brm(S ~ landscape * log.area)
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	21.74	0.92	19.96	23.54
landscapeBauxite				-5.74
landscapeForest				0.47
landscapeUrban				-5.11
log.area.				13.64
landscapeBauxite:log.area.	5.10	2.08	1.01	9.21
landscapeForest:log.area.	-0.23	2.10	-4.42	3.93
landscapeUrban:log.area.	1.86	2.60	-3.27	7.08

Summary table not
helpful for predictors
with >2 levels

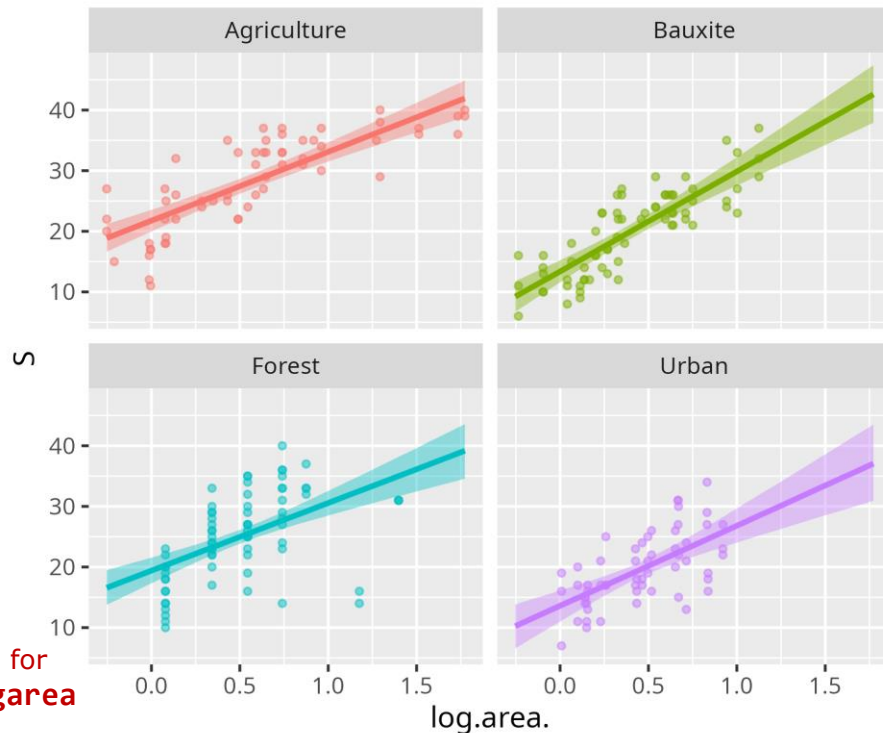
Q: Is the species-area relationship different between landscape types?

```
L00(fit_factorial, fit_additive)
```

Model comparisons:

	elpd_diff	se_diff
fit_factorial	0.0	0.0
fit_additive	-0.4	2.4

No strong evidence for
S~landscape*logarea
against
S~landscape+logarea



Interaction model

```
> brm(S ~ landscape * log.area)
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI
Intercept	21.74	0.92	19.96	23.54
landscapeBauxite				-5.74
landscapeForest				0.47
landscapeUrban				-5.11
log.area.				13.64
landscapeBauxite:log.area.	5.10	2.08	1.01	9.21
landscapeForest:log.area.	-0.23	2.10	-4.42	3.93
landscapeUrban:log.area.	1.86	2.60	-3.27	7.08

Summary table not
helpful for predictors
with >2 levels

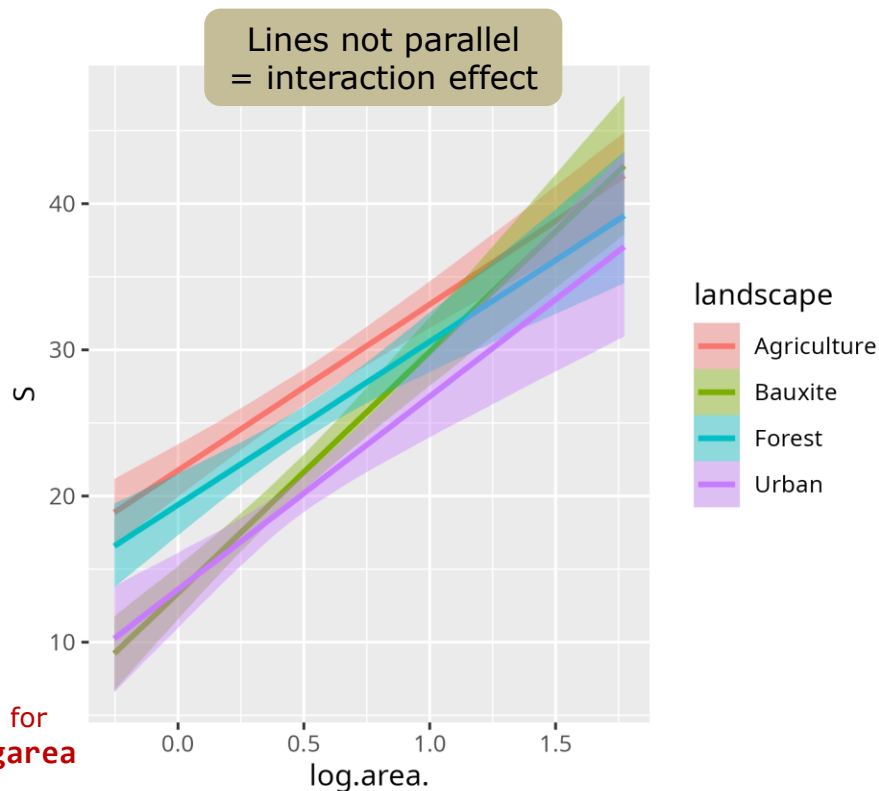
Q: Is the species-area relationship different between landscape types?

```
L00(fit_factorial, fit_additive)
```

Model comparisons:

	elpd_diff	se_diff
fit_factorial	0.0	0.0
fit_additive	-0.4	2.4

No strong evidence for
 $S \sim \text{landscape} * \log \text{area}$
against
 $S \sim \text{landscape} + \log \text{area}$



Interaction model

Post-hoc analysis

(If there was support for this model)

```
> brm(S ~ landscape * log.area)
```

Q: What are the predicted slopes?

```
> emtrends(fit_factorial, ~landscape, var="log.area.")
```

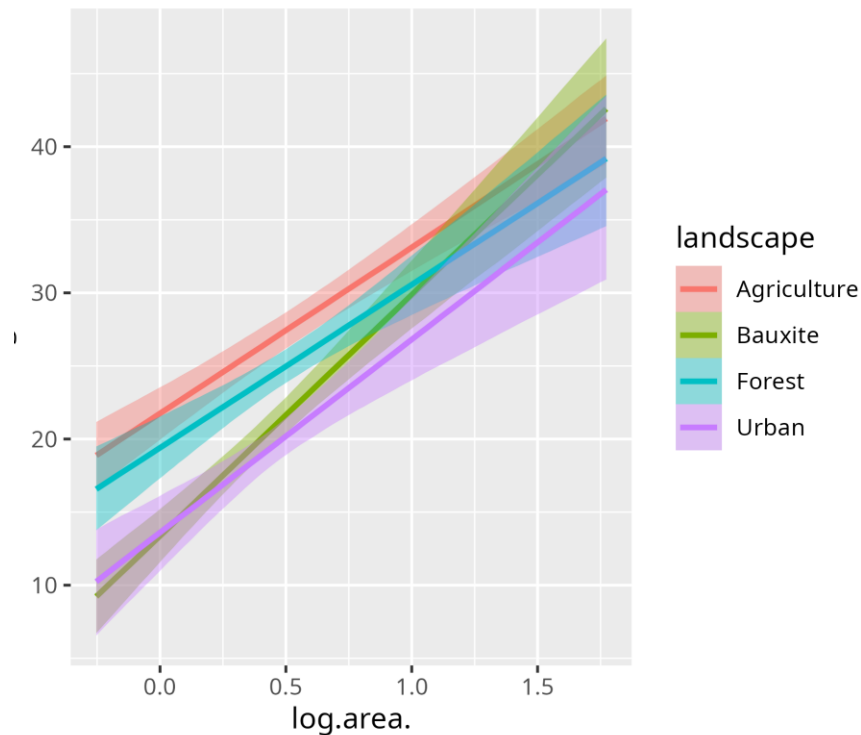
landscape	log.area..trend	lower.HPD	upper.HPD
Agriculture	11.4	9.12	13.8
Bauxite	16.5	13.26	19.9
Forest	11.1	7.81	14.7
Urban	13.2	8.77	18.1

Point estimate displayed: median

HPD interval probability: 0.95

```
> emtrends(fit_factorial, ~landscape, var="log.area.") |> pairs()
```

contrast	estimate	lower.HPD	upper.HPD
Agriculture - Bauxite	-5.066	-9.224	-1.04
Agriculture - Forest	0.224	-3.914	4.42
Agriculture - Urban	-1.867	-7.125	3.19
Bauxite - Forest	5.325	0.292	10.10
Bauxite - Urban	3.206	-2.198	9.12
Forest - Urban	-2.070	-8.073	3.42



Summary

Summary

- Regression, ANOVA, ANCOVA are just linear models
- Categorical variables can be expressed by „dummy-coding“ or by „effects-coding“, brms uses dummy-coding as default
- In Bayesian stats, linearity is not that important
- But always check your model assumptions (e.g. PPC, check_model)
- Research question should guide you which model to fit and which „tests“ to perform
- „Test“ just means a statement about a research question, quantified through posterior distribution of effect sizes, model comparisons, or post-hoc analysis
- brms flexible and „all-in-one“ package

Further reading

Bürkner, P. (2024). The brms Book [in progress]. <https://paulbuerkner.com/software/brms-book/>

Cinelli, C., Forney, A., & Pearl, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, 53(3), 1071–1104. <https://doi.org/10.1177/00491241221099552>

Conn, P. B., Johnson, D. S., Williams, P. J., Melin, S. R., & Hooten, M. B. (2018). A guide to Bayesian model checking for ecologists. *Ecological Monographs*, 88(4), 526–542. <https://doi.org/10.1002/ecm.1314>

Fieberg, J. (2024). Statistics 4 Ecologists. <https://statistics4ecologists-v2.netlify.app/> [Chapters 1,3]

Gelman, A., Hill, J., & Vehtari, A. (2020). Regression and Other Stories. *Cambridge University Press*. <https://doi.org/10.1017/9781139161879> [Chapters 6-12]

Inchausti, P. (2023). Statistical Modeling With R: a dual frequentist and Bayesian approach for life scientists. *Oxford University Press*. [Chapters 4-7]

Kery, M. & Kellner, F. (2024): Applied Statistical Modelling for Ecologists. *Elsevier*. [Chapters 5-9]